



FITTER-EU

DELIVERABLE 2.5

FITTER Community Groups of Stakeholders Database, Manual and Action Plans



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PROJECT ACRONYM	FITTER-EU
Project ref. number	101132546
Document title	FITTER Community Groups of Stakeholders Database, Manual and Action Plans
Document type	R – Document, report
Due date of deliverable	30/09/2024
Submission date	04/10/2024
Status	Final
Dissemination level	SEN – Sensitive
Language	English
Organisation responsible for deliverable	Smart Venice (SV)
Author(s)	Maria Sangiuliano, Oriana Salomón Balsamo, Anita Mezza, Tabitha Goebel
With contributions by	All partners

REVISION HISTORY			
Version	Date	Modified by	Comments
0.0	06/09/2024	Oriana Salomón Balsamo, Maria Sangiuliano	Chapter 4 shared for review by specific partners
0.1	09/09/2024	Alicja Bobek	Review of Chapter 4
0.2	10/09/2024	Sara Clavero	Review of Chapter 4
0.3	11/09/2024	Tom Flynn	Review of Chapter 4
1.0	16/09/2024	Maria Sangiuliano, Oriana Salomón Balsamo, Anita Mezza, Tabitha Goebel	Full first draft shared for review by all partners
1.1	17/09/2024	Tom Flynn	Review and additional input
1.2	Until 21/09/2024	Mehdi Zarehparast Malekzadeh, Pia Wieser, Laura Armayones Carranza, Noirin MacNamara, Olga Kolotouchkina	Review and additional input

2.0	30/09/2024	Maria Sangiuliano, Oriana Salomón Balsamo, Anita Mezza, Tabitha Goebel	Revisions integrated
3.0	04/09/2024	Sara Clavero	Final review

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List of acronyms

AI	Artificial Intelligence
AITEC	AITEC Asesores Internacionales SRL
CCGS	Cluster Community Group of Stakeholders
CRM	Customer Relationship Management
CSO	Civil society organisation
CSR	Corporate Social Responsibility
D	Deliverable
DE	Germany
DG	Disadvantaged group
EAB	External Advisory Board
ES	Spain
EU	European Union
EURECAT	Fundacio Eurecat
FGB	Fondazione Giacomo Brodolini
GA	Grant Agreement
HU	Hungary
IE	Ireland
ILO	International Labour Organisation
IoT	Internet of Things
IT	Italy
IUCN	International Union for Conservation of Nature
LGBTQIA+	Lesbian, gay, bisexual, transgender, queer, intersex and asexual people
NAP	National Action Plan
NRRP	National Recovery and Resilience Plan
PM	Project Management
PT	Portugal
QR code	Quick Response code
RRF	Recovery and Resilience Facility
RRI	Responsible Research and Innovation
SFC	Szekely Family & Co. Nonprofit korlatolt felelossegu tarsaag
SME	Small and medium enterprise
SV	Smart Venice SRL
T	Task
TFC	TFC Research and Innovation Limited
TUD	Technological University Dublin
UCM	Universidad Complutense de Madrid
UN	United Nations
UNINOVA	UNINOVA Instituto de desenvolvimento de onvas tecnologias - Associação
WECF	Women Engage for a Common Future EV
WINTA	Women, inter, non-binary, trans, agender
WP	Work package

1 Introduction

1.1 Overview of FITTER-EU

The FITTER-EU project aims to contribute to existing research on the origins, dynamics and determinants of inequalities and enable anticipatory governance to support a fair and inclusive twin transition across Europe. The project innovates through the formulation and development of an ecosystem (i.e., FITTER ecosystem) that pivotally includes a highly interactive and gamified Digital Platform powered by an Intelligent Predictive Decision Support System. Through a co-creation methodological approach, this ecosystem aims to enable policymakers to predict which social groups may be at risk of being adversely affected by twin transition policies under different scenarios, with the option to simulate the implementation of these policies for the assessment of inequalities and social exclusion risks. The Digital Platform within the ecosystem aims to provide proposed mitigation measures and better practice guides that can be used to curve potential negative effects of policies on identified at-risk groups. In addition, the FITTER ecosystem also incorporates wide platform connectivity capability and better practice guides which, through an applied foresight engagement process, helps to support the needs of the engaging stakeholders within the community of the FITTER Cluster Group. The group includes key pan-European members such as policymakers, civil society organisations including trade unions and business angels.

1.2 Purpose and scope of the Deliverable

In Work Package (WP) 2, “Conceptual framework and methodology”, the FITTER-EU project develops the conceptual and methodological framework that guides and informs the project's research. Task (T) 2.1, “Just transition definition and research design”, provided an operationalised definition of a ‘just transition’, while T2.2, Governing the twin transitions, developed a model of effective transition governance based on a mapping of structures and practices already in place in Europe. This deliverable, part of T2.3 “FITTER expert group engagement strategy” contributes to WP2’s aim of refining the research methodology and establishing a dedicated methodological approach for the engagement of stakeholders and targets from within the Cluster Community Group of Stakeholders (CCGS), as established in WP7, Dissemination, communication and outreach.

This deliverable presents the project’s CCGS Database, CCGS Manual, and six country CCGS Action Plans. The database details the stakeholders that will be contacted and involved, while the manual outlines the strategy for identifying and engaging with key stakeholders in the CCGS. The Expert Group will be consulted during various methodological steps throughout the project’s implementation. Their involvement is thoroughly described in the manual. It will serve as a living document, continuously updated throughout the project. Both the database and the manual will be used as references in all steps involving co-creation activities with the Expert Group. Lastly, Smart Venice (SV) worked jointly with each national partner to adapt the manual to the particularities of each individual country (DE, IE, ES, IT, HU and PT), define a list of prioritised actors and an engagement action plan. For countries represented by more than one partner (IE, ES and IT), a lead partner was assigned.

In terms of links with other WPs, this deliverable is inextricably connected to all WPs, and specifically with the tasks that include a stakeholder engagement component (listed in Table 1). The engagement activities foreseen in the project across these tasks include various types of workshops, focus groups, semi-

structured interviews, surveys and questionnaires, labs, policy roundtables, walkthroughs and hackathons, as well as a hybrid final conference.

Table 1: List of FITTER-EU Tasks that include a stakeholder engagement component

WP	Task
WP3 - Scenarios, hazards and vulnerability mapping	- T3.1 - Mapping national baseline scenarios and associated vulnerabilities - T3.2 - Defining future scenarios - T3.3 - Mapping twin transition hazards and negative impacts
WP4 - Inequalities risk assessment and intersectional mitigation plans	- T4.1 - Scenario-based identification and analysis of inequality risks - T4.2 - Co-creating intersectional mitigation plans
WP5 - FITTER Digital platform	- T5.1 - Platform co-design and co-creation - T5.2 - System and features design
WP6 - Validation and demonstration	- T6.1 - Validation of methods, theories and processes - T6.2 - Validation of software components - T6.3 - Pilot testing: Case-studies
WP7 - Dissemination, communication and outreach	- T7.1 - Dissemination and communication - T7.2 - FITTER Cluster Community Group Engagement - T7.4 - Exploitation, uptake and sustainability

The FITTER-EU project has six national cases, each with a partner in charge of coordinating the related research activities. However, there are three countries where more than one partner contributes:

- Germany (WECF)
- Hungary (SFC)
- Italy (FGB), contribution of SV.
- Ireland (TU Dublin), contribution of TFC.
- Portugal (UNINOVA)
- Spain (UCM), contribution of AITEC and Eurecat.

This deliverable relies on key definitions developed by D2.1. These definitions can be reviewed in Box 1.

Box 1. Project glossary from D2.1

- Green transition: “A response to climate change, which has been characterised as a ‘statistically significant variation in the mean state of the climate or in its variability, persisting for an extended period’ (IUCN, 2011) and can be caused by natural processes or human activity (Eurofound, 2023)”.
- Digital transition: “A transition from mechanical and electronic to digital electronics – a shift that started in the 1970s and included the adoption of personal computers and mobile phones and, more recently, the growth in usage of new technologies, including AI, robotics IoT (Fouquet & Hippe, 2022)”.
- Twin transition: “A process which intertwines the green and the digital transitions with a goal to consider them equally important for the future of societies (Koundouri et al., 2023, p. 3)”.

Source: FITTER-EU Project (2024).

1.3 Structure of the Deliverable

This deliverable is structured in five parts. Chapter 2 provides the methodological background underpinning this deliverable. Chapter 3 presents the FITTER-EU engagement principles, while Chapter 4 covers the project's engagement strategy of the project, proposing two paths to operationalise the engagement and procedures to be followed by partners. Chapter 5 presents challenges, mitigation strategies and incentives for engagement. Finally, Chapter 6 develops the engagement strategies per type of stakeholder targeted by the project, analysing the information developed in the national action plans (NAP) and giving guidelines per country.

2 Methodological background

As outlined in D2.1, a key component of participatory policymaking is inclusive collaboration. According to the principle of inclusive collaboration, and the systemic, complexity-oriented theoretical approach FITTER-EU is embracing, engagement strategies should acknowledge the “different kinds of knowledge [that diverse stakeholders] hold, including lived, professional and specialist expertise” (Blomkamp, 2022, p. 18). As illustrated by Bondy & Charles (2020), “organisations are not islands of activity, and must relate in some way to the social and natural systems in which they operate” (p. 68). Consultation with a wide range of stakeholders increases the possibility that policy design and implementation adopt a user-oriented approach and incorporate “a more explicit, articulated, transformative, and less-biased approach to intersectionality since the inclusion of embodied subjects expressing different concerns can promote greater self-reflexivity on one’s own biases” (Lombardo & Rolandsen Agustín, 2012, p. 491). D2.1 also emphasises that procedural justice requires that stakeholders be included in decision-making in a non-discriminatory way (McCauley et al., 2013).

This Chapter introduces stakeholder engagement and the methodological background underpinning the present deliverable. An exploratory literature review was conducted using the search string “stakeholder engagement” AND “transition” AND (“twin” OR “green” OR “digital”). In addition, a search was conducted for stakeholder engagement manuals of other Horizon Europe projects on topics related to FITTER-EU. In total, this search resulted in the identification of 40 sources which informed the development of this Chapter.

2.1 Introducing stakeholder engagement

Growing literature emphasises the importance of consultation, collaboration and dialogue with stakeholders to inform research processes and improve research uptake (Huzzard, 2021), as well as to make policymaking fairer (Bianchi et al., 2024). Griessler & Stack (2023) argue that “diversity of thoughts leads to better Research and Innovation outcomes” (p. 162). In a similar vein, Martinez et al. (2019) note that “stakeholder perspectives can guide the development of a research question and subsequent study implementation, data analysis and dissemination of study findings. As a result, study outcomes are more likely to be rapidly actionable, feasible, trustworthy and valued by the public” (p. 1327). Stakeholder engagement provides the advantage of “producing findings/solutions/policies that are, on the long-term, acknowledged by a broad variety of actors” Griessler & Stack, 2023, (p. 162).

Stakeholder engagement is discussed in the context of both research and policymaking. Additionally, in the last decade, the **Responsible Research and Innovation (RRI) framework**, where engagement and participation are key determinants of ‘responsibility’, has also been adopted in both public/academic research and in the private sector. The latter has often been framed within “Corporate Social Responsibility” initiatives, resulting in stakeholder engagement strategies for businesses (Kujala et al., 2022; Marcon Nora et al., 2023). RRI has gained prominence in the EU, partly due to the European Commission’s efforts in funding various RRI calls for proposals under Horizon 2020 (European Commission, 2013). In research, stakeholder engagement literature is particularly advanced in the field of health, in which sector-specific approaches and guidelines for successful engagement have been developed (e.g., Concannon et al., 2014; Forsythe et al., 2016; Hinchcliff et al., 2014).

Stakeholder engagement has had a central role in the field of climate change, with the 1994 *UN Framework Convention on Climate Change* calling for wide cooperation in the international response

(Jetoo, 2019, p. 2). Similarly, the International Labour Organisation (ILO) *Guidelines for a just transition towards environmentally sustainable economies and societies for all* underline the importance of “adequate, informed and ongoing consultation” taking place “with all relevant stakeholders” (ILO, 2015, p. 5). Stakeholder engagement is particularly relevant in the context of policies related to the twin transition, due to their potential to affect the environments, economies and labour market conditions of countries, regions and communities. As illustrated in D2.1 (see pp. 59-60), EU policy documents (e.g., Council of the European Union, *Silesia Declaration on Solidarity and Just Transition*) recognise the interlinked nature of climate change, inequity and poverty, and outline provisions to protect structurally vulnerable workers. These connections result in a need for social dialogue and consensus-building, because inclusive and robust solutions emerge when stakeholders actively engage in context-specific decision-making. While the term ‘stakeholder’ is characterised by diverging and sometimes unspecific definitions (Griessler & Stack, 2023; Haeberlein & Hoevel, 2023; Kujala et al., 2022), the literature on the topic converges on some crucial defining features.

Stakeholder engagement can be categorised into ‘broad’ and ‘narrow’ definitions.

Narrow definitions view stakeholders as the actual subset of stakeholders who engage with the project (Topaloglou et al., 2023), sometimes including the actual persons/groups (Forsythe et al., 2016).

Broad definitions view stakeholders as all the persons, groups or organisations who may influence or be influenced by a project (Huzzard, 2021).

In broad definitions, stakeholders are often identified on the basis on their interest in the processes and outcomes of a project or decision (Boaz et al., 2018; Gardner et al., 2009; TeRRIFICA, 2019). Definitions converge on identifying stakeholders because of their being affected by a project (Concannon et al., 2014; Mallery et al., 2012). Silvius & Schipper (2019) extend the scope of stakeholders to also cover those who perceive themselves as being affected by a project. The mutual interest or influence between a project and its stakeholders is described in many different ways. For example, Jolibert & Wesselink (2012) view stakeholders as the users of research findings, or those who benefit from them. Mallery et al. (2012) give weight to potential for involvement in decision-making, and to potential for influencing the ability of an organisation to achieve its objective and for providing input on whether the objective is achieved. Bianchi et al., (2024) summarise by describing stakeholders as those who “can help develop, implement and evaluate” (p. 17) research and policy.

A project’s stakeholders can be organised into actor categories. The subset of relevant stakeholder categories for a given project is determined by several project characteristics, such as its geographical scope, sector/field of work, and thematic focus. For example, the Europe-LAND project, aiming to develop, test and implement integrated tools for understanding factors behind land use decisions, lists the following stakeholder categories: academia, industry, policy/government, societal actors/community, and other EU projects (EuropeLAND Project, 2023).

Box 2. FITTER-EU project definition of “stakeholders”

The FITTER-EU project definition of “stakeholders” (using a broad definition) includes actors (individuals, groups or organisations) who influence and are influenced by the twin transitions, and those whose expertise (academic, practical or policy-focused) can help develop, implement and evaluate the FITTER-EU project, both as far as its research components are concerned and its innovations/IT based decision-making support system. In particular, the FITTER-EU project identifies the following stakeholder categories: civil society organisations, employer’s organisations, public service/policymaking, private sector, research/academia, trade unions and business angels. In FITTER-EU, stakeholders partly coincide with research subjects, i.e. structurally vulnerable groups affected by the twin transition.

Interactions between research projects and their stakeholders can be one-way or two-way. One-way communication concerns dissemination activities through which researchers convey project information and results, for example through publications, newsletters or videos. On the other hand, two-way communication, taking place in workshops, meetings, conferences or dedicated interactions, promotes opportunities for dialogue, exchange and the joint construction of knowledge (Jolibert & Wesselink, 2012). Even if the focus of this deliverable is on the project activities that involve two-way communication, it also integrates one-way communication actions that are part of broader project dissemination activities in WP7 (see Section 4.3), as the two dimensions are mutually supportive and functional to each other.

2.2 Stakeholder engagement: Definitions

Stakeholder engagement is an iterative process, by which researchers actively solicit “the knowledge, experience, judgement and values of individuals selected to represent a broad range of direct interests in a particular issue, for the dual purposes of: creating a shared understanding; making relevant, transparent and effective decisions” (Deverka et al., 2012, p.5 as cited in Boaz et al., 2018). The results of stakeholder engagement include informed decision-making, specifically on how to design, conduct and use research (Concannon et al., 2014). Other EU-funded projects with a focus on climate change focus on the importance of creating and maintaining relationships. For example, the TeRRIFICA project, aiming to influence climate change policies and actions by fostering Responsible Research and Innovation, asserts that “engagement is the process of building relationships with people and putting those relationships to work to accomplish shared goals” (TeRRIFICA, 2019, p.5). Jetoo, (2019) identifies stakeholder engagement as “one of the responses to the general lack of effectiveness of environmental policy” (p. 4), especially due to climate change being a wicked problem with no easy solutions.

Types of stakeholder engagement vary by intensity (frequency and duration of interactions, depth of exchanges, effort required both on the part of researchers and on the part of stakeholders, degree of involvement in decision-making) (ALLReady Project, 2022; GRIPP Project, 2020; Hering et al., 2012; QUEST Project, 2019) and the type of stakeholder engagement required depends on the objectives being pursued. For example, contentious or complex decisions will require more interactive and deliberative processes, while for simpler issues, lighter forms of engagement might be sufficient (Gardner et al., 2009). The QUEST project presents a ladder of engagement (Table 2), displaying different levels of engagement.

Table 2: A ladder of engagement, displaying different intensities of stakeholder engagement (QUEST Project, 2019).

Type of engagement	Description
Co-decision	Stakeholders not only provide several options for future actions but also decide via voting, agreeing, etc.
Co-deliberation	Concrete suggestions and future directions are suggested by participants
Dialogue	An exchange of thoughts, in the form of focus groups or democratic fora
Consultation	A systematic gathering of opinions, ideas or suggestions, for instance through surveys or other means
Information	A one-way information flow, possibly including Q&A time at the end of an information meeting

Overall, stakeholder engagement denotes a collaborative effort (Gardner et al., 2009), which should be ongoing (Bianchi et al., 2024; Hering et al., 2012), and include moral, strategic and pragmatic considerations (Haeberlein & Hoevel, 2023; Kujala et al., 2022).

Given its scope and theoretical approach **FITTER-EU, as anticipated, undertakes an approach to stakeholder engagement that is inspired by intersectionality theory/practice**. This entails paying attention to how power dynamics and existing structural inequalities featuring a given context affect or could affect engagement efforts (Søraa et al., 2020b), leading to excluding certain (less powerful/visible and/or more critical/radical) stakeholders or ‘silencing’ them once engaged (Bondy & Charles, 2020). There is limited specific literature on applying intersectionality to stakeholder engagement as such, while inspiration can be taken by efforts made in close fields such as policy evaluation. For example, the ISE4GEM approach to policy evaluation uses interesting frames and tools for stakeholder analysis that try to anticipate intersectional vulnerabilities, if they can engage freely, or need special accommodation to participate, but also identifying ‘gatekeepers’ that could facilitate or create barriers to other stakeholders (Stephens et al., 2018). A call for those in charge of orchestrating the engagement is also made to ensure that facilitators are aware of their own “intersectional positionality” in the process and the power leverages/limitations that feature their role. These considerations will be informing Chapter 3 on engagement principles.

2.3 The stages of stakeholder engagement

Stakeholder engagement covers a broad range of activities (Boaz et al., 2018), as stakeholders can be engaged throughout the research process (Forsythe et al., 2016). Several sources list the steps of stakeholder engagement. The steps usually proposed in the literature, leading up to recruitment, include the following (Haeberlein & Hoevel, 2023; see Table 3).

Table 3: Initial steps of stakeholder engagement (Haeberlein & Hoevel, 2023, pp. 41–44).

Step	Description
Identify	Begin by brainstorming a diverse list of actors. Identify relevant stakeholders by reviewing your own or your institution's network, as well as relevant literature. Once you have an initial list of stakeholders, ask them to recommend additional potential stakeholders using their own networks and contacts (also noted in (Lim et al., 2004).
Analyse	Assign a stakeholder category and assess how each stakeholder can contribute to the project.
Map	Visualise the expertise of stakeholders in relation to their interest/willingness to participate in the project.
Prioritise	Establish the degree of potential engagement for each stakeholder type (see Table 2).
Select	Identify the best stakeholders to be contacted for specific project activities.
Recruit	Contact the selected stakeholders.

Mallery et al., (2012) provide insights on further stages, once the stakeholders have been recruited (Table 4).

Table 4: Further steps of stakeholder engagement (adapted from Mallery et al., 2012, pp. 9–15).

Step	Description
Build trust	Provide stakeholders with materials to foster meaningful participation, for instance by crafting personalised communications, listing incentives, answering questions and listening to concerns.
Offer education	Identify and address stakeholder needs for additional informational resources or training to ensure meaningful contribution.
Facilitate the process	Ensure that facilitators are skilled and neutral, and foster a safe environment.
Idea generation and prioritisation	Conduct the engagement activity so as to encourage the sharing of ideas, opinions and contributions. Record the outcomes of activities for later analysis and prioritisation, for instance using online platforms to facilitate the process.
Dissemination	Stakeholders can be involved in dissemination activities. On the other hand, keeping the engaged stakeholders informed on project's development and outcomes and how the results of the activities they have been part of have been used, put dissemination at the service of accountability towards stakeholders.

In the FITTER-EU project, the stages of stakeholder engagement are defined by the NAPs and elaborated upon in this deliverable.

Findings from the literature suggest that exchange between actors results in authentic co-production of research (Hinchcliff et al., 2014). Stakeholder engagement carries considerable benefits, from strengthening the credibility of research results (Hinchcliff et al., 2014) to bringing societal and democratic benefit (Griessler & Stack, 2023). To ensure the achievement of these outcomes, meaningful partnerships are to be sought, rather than “engaging stakeholders merely to fulfil a requirement” (Forsythe et al., 2016, p.17), and project information should be tailored to each stakeholder (Hinchcliff et al., 2014).

3 FITTER-EU Principles for stakeholder engagement

This chapter draws on Chapter 2 to distil principles that should guide both the engagement process/strategy and facilitation practices within FITTER-EU. It is therefore important that partners' team members actively involved in stakeholder engagement, both researchers and facilitators, are aware of the approaches and guidance presented below.

3.1 Theoretical context and general principles

D2.1 sets the context for the project's stakeholder engagement strategy by elucidating key concepts. For example, it introduces the concept of 'procedural justice': "the requirement that all stakeholders be included in a non-discriminatory way; that they be able to participate in decision-making; and that government and industry stakeholders commit to participation, impartiality and full information disclosure" (FITTER-EU D2.1, p. 78). D2.1 also proposes six key principles of the FITTER System Design Practice, some of which are relevant to stakeholder engagement (see Box 2).

Box 2. Principles of the FITTER System Design Practice

- 1) **Purpose-driven** – we seek to enable a just twin transition which addresses intersectional inequalities, with fair burden-sharing, while controlling climate change and taking the dimensions of environmental, climate, and energy justice into account.
- 2) **Recognising complexity** – we commit to understanding justice concerns as situated within the 'systems trilemma'. We commit to understanding inequalities as co-constitutive (not multiple), making dynamics of oppression and privilege explicit and articulated; gendering policies; challenging privilege (Lombardo and Agustín, 2012).
- 3) **Rights-based/Self-determination** – we commit to upholding human rights, taking the core tenets of energy, environmental and climate justice into account.
- 4) **Addressing power relations** – we commit to taking a conflict resolution approach to deliberative mechanisms, acknowledging operative power dynamics, and seeking to create a negotiated compromise that does not sacrifice social justice requirements in order to achieve environmental and economic goals. We also aim to challenge harmful exclusionary norms and extend the boundaries of what is 'normal' (addressing invisible power).
- 5) **Inclusive collaboration** – we commit to consulting a wide range of stakeholders in order to adopt a user-oriented approach and incorporate 'a more explicit, articulated, transformative, and less-biased approach to intersectionality since the inclusion of embodied subjects expressing different concerns can promote greater self-reflexivity on one's own biases' (Lombardo and Agustín, 2012: 491).
- 6) **Adaptive learning** – we commit to collaborative problem-solving, openness to reframing problems, the unlearning of exclusionary norms, understanding inequalities as systemic, and ongoing adaptation throughout the design process.

To note, these principles can be further developed at team level and in consultation with key stakeholders.

Principle 5, on inclusive collaboration, is especially relevant to the project's stakeholder engagement strategy. To provide further insights into how inclusive collaboration can be achieved, this Chapter offers principles for engagement, drawing from feminist and critical realist approaches to stakeholder engagement.

Feminist critiques of traditional approaches to stakeholder engagement have pointed out that engagement, especially in the case of private sector organisations conducting stakeholder engagement, tends to benefit powerful actors, while perpetuating the marginalisation of those who are less powerful

(Bondy & Charles, 2020). This is because, as Bondy & Charles (2020) illustrate, stakeholders are selected primarily based on their power and legitimacy. Erikainen et al. (2021) elaborate on this by presenting the typical stakeholder in health-related research, the epistemically privileged conventional ‘expert’: “systemic structural inequalities and hierarchies delimit the practical realities of who is included in (and excluded from) engagements” (p. 11). They use this example to show that assumptions are often made about what ‘knowledge’ is, and who is allowed to count as a ‘knowledge producer’. However, there is now growing awareness about the importance of involving other actors in research, who “possess a particular expertise: positional experience and ability to navigate the engagements environment and effectively provide ‘lay perspectives’” (Erikainen et al., 2021, p.8). As Rouse & Woolnough (2018) explain, a critical realist approach values “creating knowledge from inside social struggles” (p. 441).

Adopting a critical realist perspective in stakeholder engagement means to seek an awareness of power hierarchies, and to take accountability for creating more empowering conditions for marginalised individuals and groups to express their views (Rouse & Woolnough, 2018). This can be achieved, for instance, through participatory action research, and actively trying to challenge and change social relations of domination. Three key choices can facilitate a feminist and critical realist approach in stakeholder engagement: rejecting essentialism, adopting reflexivity, and seeking transnational solidarities.

Bondy & Charles (2020) explain that essentialism is a component of traditional stakeholder engagement approaches, whereby, with reference to hegemonic masculinity practices, stakeholders are inevitably “othered” and understood through a crystallisation of those characteristics that emerge as salient. Stemming from enlightenment thinking, the essentialist self is gendered and embodies gendered characteristics, which are seen as natural, unchangeable ‘facts’, and enable the marginalisation of those with poorly valued characteristics. Going beyond essentialism is key in an inclusive engagement strategy, in particular by adopting a relational understanding of the self: “each self is composed of multiple, intersecting selves, that while different and sometimes in conflict, do not privilege one over another, and are constantly shaped and challenged through social relations and personal introspection within relevant contexts” (Bondy & Charles, 2020, p. 73). Different dimensions of who the stakeholder is must be taken into account, allowing for membership of more than one ‘group’. This also implies an attention to avoid tokenist approaches when addressing inclusiveness, i.e. considering having the issue covered by one participant (the ‘token’) representing disadvantaged groups and/or not integrating intersectionality in the way the contents are presented and/or the methodology unfolded.

Reflexivity encompasses a self-awareness of positionality, both on the part of the organisation and on the part of stakeholders themselves. According to Erikainen et al. (2021), “feminist reflexivity can enable us to interrogate critically the epistemic starting points and presumptions that shape how engagements are conceptualised and undertaken, and how these starting points direct the kinds of knowledges that can (not) be produced through them, especially within institutional confines” (p. 11). By adopting a reflexive approach and developing an awareness of positionality, the relationship between engagements and knowledge production can be better explored and understood, allowing researchers to learn from a plurality of knowledges (Erikainen et al., 2021).

Lastly, in presenting activist scholarship, Rouse & Woolnough (2018) emphasise the importance of understanding and recognising the global dimension of gender and other struggles, suggesting that “scholarship and activism must be networked transnationally” (p. 441). They advocate for the importance of building transnational collaborations countering the perpetuation of patterns of inequality, with a focus on carrying out research with activists (“people who have already taken the leap from outrage to action

and are ready to reflect more deeply”, p. 440) as well as “the unorganised, vulnerable and unwaged” (p. 440). This way, researchers take responsibility for their research process.

Practical recommendations to go towards more inclusive stakeholder engagement include being “much more passive in identifying stakeholders, and instead [being] active in creating support structures for stakeholder self-identification” (Bondy & Charles, 2020, p. 79). Feminist approaches to stakeholder engagement also cover evaluation: methods should be multi-vocal, focus on social inequalities, and ensure that actors are empowered to express their views (Erikainen et al., 2021).

3.2 Operationalising the principles in practice

Non-hierarchical dynamics: horizontal and participatory dynamics are encouraged between organisers/facilitators and participants. Since participants attend these events on a pro-bono basis, FITTER-EU partners should ensure not to overburden activists and participants, while recognising and valuing the time they contribute. The administrative, management, facilitation, and follow-up work should be carried out by organisers. During meetings, special care is taken to address various power dynamics that might impact facilitation and interaction among participants. The aim is to avoid the dominance of any single perspective and to create safe, inclusive spaces for everyone. Feminist pedagogies, which are grounded in standpoint and positionality theories, inform this facilitation approach. This involves self-reflective acknowledgment of one’s own position in relation to privilege and power structures, considering multiple axes of difference and inequality (e.g., age, class, ethnicity, sexual orientation, gender identity, abilities, etc.).

Different forms of expression and knowledge sharing: particular care should be taken when participants from structurally vulnerable or disadvantaged groups, who may have limited experience in these types of research and innovation settings, are involved. Facilitators should provide space for diverse ways of knowledge-sharing and expression, beyond just analytical and rational arguments. This can include personal life stories or accounts of how policies affect them, as well as narratives of resilience and resistance at both individual and collective/political levels.

Facilitating participation for those with limited resources

When participation fees are not available, explore alternative forms of compensation your organisation can offer (e.g., vouchers, meals during events, coverage of transport costs, or access to libraries or other cultural/leisure activities). Additionally, it’s important to accommodate participants with caregiving responsibilities—often women or less affluent individuals who may not have the means to pay for care services. This can be addressed by scheduling meetings during regular daytime hours, avoiding early mornings, late evenings, or weekends, or by providing childcare services, either through on-campus facilities or other available options.

Decision-making

Decision-making dynamics can be understood using Kaner’s “diamond of participation,” which describes how collaborative groups progress through various developmental stages (White, Hunter, and Greaves 2022):

1. **Discussion:** A new topic is introduced.
2. **Divergent thinking:** New ideas and perspectives are explored. An open and respectful environment is essential, where no idea is judged or dismissed prematurely.

3. **Groan zone:** This phase is characterised by conflict, confusion, and ambiguity. Frustration and defensiveness are common, but this stage often leads to new understandings.
4. **Convergent thinking:** Priorities are set, agreements are reached, or new ideas emerge that help to lead to a decision.

If the specific FITTER-EU methodology includes a deliberative component, achieving universal consensus cannot be the only option. Instead, a partial agreement can be acceptable. Arguments can be discussed iteratively until common ground is found between opposing viewpoints.

Openness and Transparency:

- Clearly communicate all aspects of the event(s) participants are invited to, including the “Terms and Conditions of Participation.” This ensures clarity regarding how shared or collected information will be handled, how personal and sensitive data will be protected, and how the event will contribute to the Research and Innovation process within FITTER-EU. Consider participants’ expectations and interests, as expressed during the invitation or engagement process, or those that can be reasonably assumed based on previous interactions.
- Provide all relevant information and materials about the event well in advance, allowing participants sufficient time to understand the content being presented or discussed, as well as what is expected from them. This also enables them to organise their schedules accordingly.

Accountability:

- FITTER-EU partners should offer participants follow-up information on the progress of the engagement process and the project, while actively encouraging feedback from participants.
- If further engagement is anticipated, clearly communicate the next steps to ensure participants remain informed.

4 Overall FITTER-EU Strategy and Engagement Plan

This chapter presents the overall FITTER-EU strategy and engagement plan, grounded in the methodological background and the principles outlined in Chapters 2 and 3, respectively. The first two sections provide an overview of the engagement structures in the project and the needs determined by the project scope and participatory approach. Following this, a dual engagement strategy is introduced, encompassing both “top-down” and “bottom-up” paths. Finally, the last Section addresses specific procedures for stakeholder engagement, including guidelines on when to start it, processes to follow when a task foresees a participatory activity, coordination of partners contributing to the same country-study, general recommendations for preparing participatory activities, and guidelines for engaging stakeholders targeting the transnational/European levels.

4.1 Engagement structures throughout the project

The FITTER-EU project studies twin transitions and their effects on population groups, contributing to research on inequality, its origins, dynamics, and determinants. Specifically, the project analyses policies at different levels of governance in Europe, their present and future effects, and forecasts future scenarios in the four prioritised sectors of the project: energy, building & housing, transport & mobility, and food & agriculture. Additionally, the project proposes mitigation measures and plans—policy proposals to reduce inequality effects to tackle the negative effects of the transitions — with the objective of supporting policymakers in decision-making. To this end, a recommender system will be developed as part of the project.

To achieve its objectives, FITTER-EU is implemented with a mixed-methods methodology based on co-creation principles, making it highly participatory. Specifically, it involves the creation of the FITTER-EU Cluster Community Group (CCGS), a diverse group of stakeholders—policymakers, external experts, civil society organisations (CSOs), research and academia experts, among others—engaged from the beginning of the project to participate in the different steps of the methodology. As established in the Grant Agreement (GA): “This becomes an iterative process in the entire methodology until an appropriate or a fail-safe correction of policies is reached. Therefore, FITTER becomes an entity where multiple stakeholders come together through the CCGS to co-create policies with a view to realising the EU's goal of a twin transition process that is fair and inclusive for all.” (p. 8, part B)

To ensure the smooth implementation of all methodological steps to reach the project’s objectives, the FITTER-EU engagement strategy is emphasised and developed in three dedicated tasks, led by Smart Venice and the Instituto Desenvolvimento de Novas Tecnologias - UNINOVA, project partners from Italy and Portugal, respectively. The three specific tasks focused on engagement are outlined in Table 5.

Table 5: FITTER-EU engagement tasks

Task	Task leader	Time	Description of the task
2.3: FITTER-EU expert group engagement strategy	Smart Venice	M04-M21	This task establishes an expert group of national and transnational stakeholders to be consulted in the participatory steps of the FITTER-EU methodology, relevant according to their scholarly or practical expertise. It includes the development of a manual for engagement, a periodically updated stakeholder database, and national action plans.
2.4: Engagement	Smart Venice	M12-M14	Identification and involvement of disadvantaged groups (DGs) negatively affected by the twin transition effects in the four

Task	Task leader	Time	Description of the task
strategy for disadvantaged/vulnerable groups			FITTER-EU sectors, according to the input of T3.3. These stakeholders will participate in the labs of T4.2.
7.2: Cluster Community Group Engagement	UNINOVA	M01-M36	This task focuses on building the CCGS, identifying and involving a set of EU, transnational, and national stakeholders. Activities include: a) establishing the CCGS based on expressions of interest, b) supporting T2.3's engagement strategy beyond the six country cases, c) organising workshops and collective intelligence sessions to explore the societal, technological, and business potential of the FITTER-EU ecosystem, d) monitoring stakeholders' perceptions of inequalities, e) engaging with related groups to address challenges identified in WP3, f) developing communication strategies with the general public.

The FITTER-EU project targets different stakeholder categories, which form the basis for developing the stakeholder engagement strategy: (i) CSOs, (ii) employers' organisations, (iii) public service/policymaking, (iv) private sector, (v) research/academia, (vi) trade unions, and (vii) business angels. See Chapter 6 for specific engagement strategies per stakeholder category. Their participation in the FITTER-EU methodology interweaves in different types of activities according to their relevance for the project tasks. The project's GA describes the initial project conceptualisation and methodological proposal, detailing several stakeholder structures. Since some overlaps exist in the definitions included in the GA, it is worth clarifying the initial definitions, as well as the shape and approach they have taken after the initial months of project implementation, drawing upon the rich discussions on the matter held among partners in the first year's Project Meetings.

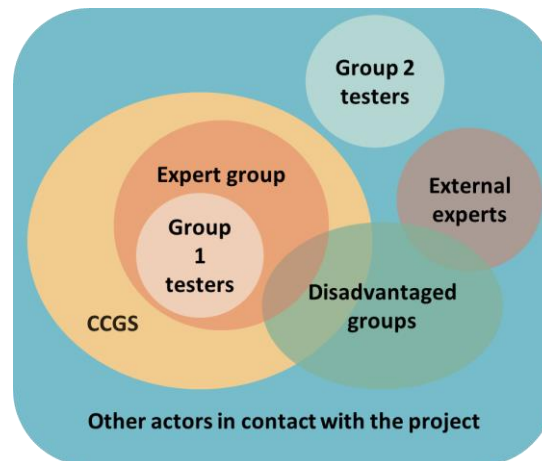
4.1.1 Interactions among the stakeholder structures in the FITTER-EU project

In March 2024, Smart Venice started discussions with UNINOVA and TU Dublin about the synergies between T2.3 and T7.2, as well as the definitions of the CCGS and the interactions between the previously presented stakeholder structures as defined in the GA. Based on these exchanges, during the monthly project meeting on April 2nd, two options were presented to the whole project consortium:

- Option 1: The CCGS as a completely open group of stakeholders belonging to the seven categories targeted by the project—CSOs, employers' organisations, public service/policymaking, private sector, research/academia, trade unions, and business angels. This option implies an overlap with the stakeholders' 'living' database initially built in T2.3, which grows organically through the project in line with database updates.
- Option 2: The CCGS as a "membership-based" group in which stakeholders explicitly declare their interest in being part of the FITTER-EU project by filling out dedicated "Terms and Conditions". In this option, there is continuity in the CCGS, with a partially fixed group of stakeholders who are periodically involved in the participatory activities of the project. The CCGS remains open and can be expanded throughout the project by following a selection process (to be defined).

After these initial exchanges, no explicit decision was made at the consortium level regarding the aforementioned options. However, considering the project developments, it is possible to state that the CCGS has taken on the characteristics of Option 1. In line with this, the following figure presents the synergies between the existing stakeholder structures in the project.

Figure 1: Interactions between the stakeholder structures of the FITTER-EU project



Below some remarks regarding the interactions between the stakeholder structures of the project are detailed:

- The **CCGS** corresponds to the FITTER-EU stakeholder database; thus, the CCGS comprises all potential stakeholders mapped as relevant for involvement in the FITTER-EU project through the database. Stakeholders could be included in the CCGS either top-down (mapped by project partners, included in the database and invited) or bottom-up (self-declaring their interest to participate in the project by filling out an online form on the project's website that 'interacts' /feeds into the database. See Section 4.3 for more details on the proposed dual engagement strategy paths of the FITTER-EU project.
- The **Expert Group** consists of stakeholders contributing to the FITTER-EU methodology with their expertise—based on their scholarly or practical knowledge—by participating in the project activities. Thus, the expert group comprises stakeholders from the CCGS contacted and who accept to participate in the project activities. Within the database, they will specifically be marked as Expert Group members once they participate in one or more activities. Continuity in participation will not be either asked for or promised by default, unless partners identify some of the stakeholders from this group as strategic for their national engagement strategies.
- **Group 1 testers** are a subgroup of the expert group and, consequently, also a subgroup of the CCGS. In contrast, **Group 2 testers** are stakeholders who collaborate with the FITTER-EU project on a one-time basis and are therefore not part of the CCGS. The latter will be included in the database only at a later stage and specifically marked as Group 2 Testers.
- Similarly to the Group 2 of testers, **external experts** are stakeholders who might be in contact with the project for a specific and limited contribution such as an informative workshop to present project results and get particular feedback. Since they do not collaborate in a more stable way to the FITTER-EU methodology and project tasks, they are not part of the CCGS.
- **DGs** are all the stakeholders negatively affected by the twin transitions. Some of these population groups will be prioritised in the project (in T3.3) and will be part of the CCGS either in their individual capacity (in case for example, they have gained some visibility through awareness raising actions or interventions on the media) or as representatives of organisations/movements. In case they are contacted and accept to participate in the project activities, they would become part of the expert group. Additionally, other not-prioritised DGs —not belonging to the CCGS— might be

contacted by the project for a concrete and specific contribution. In this case, they would be part of the “external experts” group.

4.2 Engagement needs in the project

Engagement activities vary according to the level of involvement with stakeholders. From the lowest to the highest intensity of engagement, participatory activities can be classified as information, consultation, dialogue, co-deliberation, and co-decision (see Chapter 2).

The FITTER-EU methodology includes participatory activities in all the project work packages (WPs).

WP3 participatory activities focus on gathering inputs regarding the research components of the project, obtaining feedback from stakeholders on the project’s conceptual and methodological approach, including forecasts of future scenarios, and the (in)equality effects of the twin transitions. Participatory activities in this WP are closer to information and consultation activities, lighter forms of engagement where stakeholders contribute by providing their knowledge and/or expertise on specific aspects.

WP4 encompasses the project activities with the highest level of participation. Specifically, labs implemented in WP4 include five encounters where mitigation measures to overcome the negative impacts of twin transitions are discussed and agreed upon jointly with the lab participants. This type of participatory activity is closer to co-deliberation and co-decision methods, as lab members have a central and active role in the decision-making processes, directly proposing concrete solutions and future directions.

WP5 and WP6 activities, dedicated to the co-creation and validation of the FITTER-EU platform and software components, also involve stakeholders more intensely.

In particular, FITTER-EU project activities can be clustered as follows:

- **Cluster 1: Individual engagement activities for the development of the FITTER ecosystem**

The project includes two types of individual engagement activities: interviews and surveys. The former are present in tasks 3.1 and 3.3 as semi-structured interviews, dedicated to investigating the nature of vulnerabilities and DGs, respectively. Surveys are present in task 4.1 to calculate probability rates and impact severity rates of hazards—negative impacts of the twin transitions. In addition, task 3.2 contemplates the implementation of one Delphi survey.

- **Cluster 2: Group engagement activities for the development of the FITTER ecosystem**

These are the most frequent type of engagement activities present throughout the FITTER-EU methodology, in different modalities: workshops (T3.1, T3.2, T5.1, T6.1), focus groups (T3.1, T3.2, T3.3, T5.1, T5.2), and labs (T4.2). These participatory activities, mostly held online, are closely related to the project’s methodological tasks, which determine their objectives and content. For example, workshops and focus groups in T3.1 are dedicated to collecting input regarding the vulnerability dimensions considered in the project. In T3.2, inputs are gathered to forecast the three scenarios of the project, and in T3.3, the focus is on defining the hazards—negative effects—of the transitions. Labs in T4.2, one of the few in-person engagement activities, focus on defining mitigation measures at the national and transnational levels. Activities in WP5 are dedicated to designing the platform and some system and design features. Some of these collective engagement activities are mixed with other methodologies, such as serious games (in T5.1).

- **Cluster 3: Sessions dedicated to validation of the FITTER-EU recommender system**

Cluster 3 groups the walkthrough methodology sessions, all conducted online. In T6.2, these sessions aim to get feedback on the use of the FITTER-EU recommender system in terms of fairness and usability, concluding with the design of the platform. T6.3 includes final piloting sessions in pan-European, online, cognitive walkthrough sessions.

- **Cluster 4: Technological demonstrations with usability assessments**

Cluster 4 includes two hackathons implemented in T6.2, dedicated to testing the FITTER-EU platform and obtaining feedback on its usability. They will be organised online with IT personnel, actual end-users from practitioner organisations, and students (undergraduates/postgraduates) who will receive future scenarios developed within the project and be asked to perform impact assessments and make policy recommendations based on the system's results.

- **Cluster 5: Engagement activities dedicated to dissemination, communication and exploitation.**

Participatory activities in Cluster 5 consist of public events organised for dissemination, communication and exploitation purposes. Specifically, they include two policy roundtables defined in T7.2, organised in M15 and M35, back-to-back with the final conference (T7.2). Task 7.4, focused on exploitation and sustainability, includes the use of interviews and questionnaires to identify a business model beyond the project lifecycle.

The two tables below (Table 6 and Table 7): (i) summarise the project's needs in terms of engagement, detailing all participatory activities per task, and (ii) provide a timeline summarising the months and types of participatory activities present in them. A first version of the table was established with the information included in the GA, which was then shared with partners in April 2024 to be adjusted according to their planned activities. Partners were asked to fill columns regarding the involvement of European stakeholders and the planned implementation months. Regarding the latter, it is worth noting that most of them are considered in time ranges since not all activities have been defined by task leaders.

4.2.1 FITTER-EU project's engagement activities

Table 6: Summary of FITTER-EU participatory activities

N	Task	Task lead	Cluster of activities	Engagement activity	Stakeholders	Number of stakeholders	Level of stakeholders involved	Does the activity include transnational stakeholders ? Y/N	Format	Purpose	Month
1	3.1	UCM	Cluster 2	Workshop	CCGS members	Max 15	National/European	Y	Online	Obtain feedback on the approach for the vulnerability mapping	8
2	3.1	UCM	Cluster 2	Thematic focus groups (2 per country)	CSOs representing vulnerable groups, academic experts and public authorities	Max 10 per group	National/Regional	N	Online/in person	To obtain detailed vision of each national context of vulnerabilities and to gain insights for the semi-structured interviews with members of vulnerable communities	10
3	3.1	UCM	Cluster 1	Semi-structured interviews	Representatives of vulnerable groups	Max 60	National/Regional/Local	N	Online/In person	To obtain insights on the multi-dimensional perspective of vulnerabilities	11
4	3.1	UCM	Cluster 2	Workshop	CCGS members	Max 15	National /European	Y	Online	Disseminate and discuss D3.1	12
5	3.2	EURECAT	Cluster 2	Online focus group	Selection of CCGS experts	Max 12	National/European	Y	Online	Define 3 future scenarios within the sectors of analysis	From 10
6	3.2	EURECAT	Cluster 1	Delphi survey	Experts (CCGS, snowballing and external experts)	Approximately 50	Local/Regional/National	Y	Online	Forecast practices in the 4 main sectors of analysis from experts	8 to 11

N	Task	Task lead	Cluster of activities	Engagement activity	Stakeholders	Number of stakeholders	Level of stakeholders involved	Does the activity include transnational stakeholders ? Y/N	Format	Purpose	Month
7	3.2	EURECAT	Cluster 2	Co-creation workshop	CCGS experts	Max 12	National/European	Y	Online	Co-create and create assumptions on the 3 future scenarios within the sectors of analysis	12
8	5.1	EURECAT	Cluster 2	Online/ hybrid workshops, including a survey and serious games	Group 1 stakeholders	6 per workshop	Local/Regional/National	N	Online or hybrid	Detecting, understanding and tracking changes in the user pains, environment, regulatory framework, documents currently used for policy and regulation impact assessment and retrospect evaluations on regulations and policies	12 to 18

N	Task	Task lead	Cluster of activities	Engagement activity	Stakeholders	Number of stakeholders	Level of stakeholders involved	Does the activity include transnational stakeholders ? Y/N	Format	Purpose	Month
10	5.1	EURECAT	Cluster 2	Online focus group	National policymakers and the Regulatory Scrutiny Board of the EC	8	National/European	Y	Online	Detecting, understanding and tracking changes in the user pains, environment, regulatory framework, documents currently used for policy and regulation impact assessment and retrospect evaluations on regulations and policies	10 to 18
11	5.1	EURECAT	Cluster 4	Co-design workshops at the national level for the usability of the platform	Twin transition facilitators and stakeholders	5-10 per workshop	Local/Regional/National	N	Online	Include the end-users in the design of the platform (E.g., collect preferences, adapt functionalities, etc.)	12 to 18
12	5.2	EURECAT	Cluster 2	Focus group	Stakeholders engaged in T3.1 and T3.2	4-6	Local/Regional/National	N	Online	Define the scope of the system and the main features of the FITTER-EU Digital Platform,	tbd with AITEC

N	Task	Task lead	Cluster of activities	Engagement activity	Stakeholders	Number of stakeholders	Level of stakeholders involved	Does the activity include transnational stakeholders ? Y/N	Format	Purpose	Month
13	6.1	SFC	Cluster 2	Online workshops (tabletop exercises)	Group 1 stakeholders	tbd			Online	Benchmarking elaborated methods and get feedback on theories and processes. Assess proposed mitigation measures using the four A's approach (availability, accessibility, affordability and acceptability) to minimise cognitive and social biases in them.	tbd
14	3.3	EURECAT	Cluster 1	Semi structured interviews (5 per country)	Experts recruited through the CCGS	Max 30	Local/Regional/National	N	Online	Identification of DGs within the 4 sectors	11 to 13
15	3.3	EURECAT	Cluster 2	Online transnational focus group	Selected CCGS experts	Max 10	Local/Regional/National	N	Online	Identify hazards that can arise in relation to the 3 scenarios and vulnerability dimensions	15
16	3.3	EURECAT	Cluster 1	Semi-structured interviews	DGs in each of the 4 sectors	8	Local	N	Online	Collect data to inform mitigation	16-17

N	Task	Task lead	Cluster of activities	Engagement activity	Stakeholders	Number of stakeholders	Level of stakeholders involved	Does the activity include transnational stakeholders ? Y/N	Format	Purpose	Month
										plans with DG groups	
17	4.1	SV	Cluster 1	Survey	DGs	tbd	?	N	Online?	Calculate the probability rates and impact severity rates of hazards using quantitative data on inequality indicators	Task from 14 to 20 - TBD
18	4.2	SV	Cluster 2	National FITTER-EU labs (1 per country)	External participants from DGs and the CCGS	15 to 20 per lab	Local/Regional/National	N	In-person	Analyse challenges and solutions, proposing 1 or more mitigation measures for each one of the negative effects identified of the twin transitions (hazards).	Task from 14 to 20 - TBD

N	Task	Task lead	Cluster of activities	Engagement activity	Stakeholders	Number of stakeholders	Level of stakeholders involved	Does the activity include transnational stakeholders ? Y/N	Format	Purpose	Month
19	4.2	SV	Cluster 2	Online transnational FITTER-EU lab	Participants selected from national labs - both CCGS and DGs	30	Local/Regional/National/EU level	Y	Online	Develop transnational mitigation measures	Task from 14 to 20 - TBD
20	7.4	UNINOVA	Cluster 5	Interviews and questionnaires	CCGS					Identify a business model beyond the project lifecycle	tbd
22	7.2	UNINOVA	Cluster 5	Policy roundtables	External stakeholders		tbd		Hybrid	Discuss Policy Briefs (D7.8 and D7.9)	15 & 35
23	6.2	AITEC	Cluster 3	Online walkthrough through the Alpha-version	Group 1 testers	The same as Task 5.1 for Co-creation	Local / Regional / National	Y	Online	Gather feedback about the FITTER-EU platform (Alpha-version) and its outputs regarding fairness	30-32

N	Task	Task lead	Cluster of activities	Engagement activity	Stakeholders	Number of stakeholders	Level of stakeholders involved	Does the activity include transnational stakeholders ? Y/N	Format	Purpose	Month
24	6.2	AITEC	Cluster 4	Hackathons on the initial Closed Beta version	Group 2 of testers (IT personnel, end users from practitioner organisations (e.g., EC RSB, EP Common Approach /IAH, national scrutiny officer in MS ministries), political science/law PhD students)	15 to 20 per hackathon (2)	Local / Regional / National	Y	Online	Gather feedback about the FITTER-EU platform (initial Beta-version) in terms of usability	32-34
25	6.2	AITEC	Cluster 3	Online walkthrough through the Final Closed Beta version	Stakeholders from Group 1 + Group2	The same as Task 5.1 for Co-creation (Group 1 of Testers)	Local / Regional / National	Y	Online	Gather feedback on the FITTER-EU platform (final BETA version) in terms of usability and added value.	34
26	6.3	SV	Cluster 3	EU-wide cognitive walkthrough online sessions	Policymakers from Group 2	4 sessions	Tbd	N	Online	Perform the validation of the FITTER-EU environment in terms of usability, i.e., the potential of use by policymakers in	35?

N	Task	Task lead	Cluster of activities	Engagement activity	Stakeholders	Number of stakeholders	Level of stakeholders involved	Does the activity include transnational stakeholders ? Y/N	Format	Purpose	Month
										their decision-making processes.	
27	7.1	UNINOVA & TFC	Cluster 5	Hybrid Final Conference	All stakeholder groups relevant for the project, sister projects, general public, EC actors				Hybrid	Present the final results of the project	35

4.2.2 Chronogram of participatory activities

Table 7 below provides a tentative timeline of the project participatory tasks is presented, elaborated with information from the GA and feedback obtained from partners in April 2024. It should be noted that most dates are still considered in time ranges since task leaders have not decided yet on all the specific details on these activities. Tasks 5.2 and 6.1 have not been included because their specific dates of implementation are still to be defined.

Table 7: Timeline of FITTER-EU participatory activities

Legend: Clusters of activities

	C1: Individual engagement activities for the development of the FITTER ecosystem.
	C2: Group engagement activities for the development of the FITTER ecosystem.
	C3: Sessions dedicated to validation of the FITTER-EU recommender system.
	C4: Technological demonstrations with usability assessments.
	C5: Engagement activities dedicated to dissemination, communication and exploitation.

[illegible]

4.3 A dual strategy: Engagement paths in the FITTER-EU project

The proposed engagement strategy for the FITTER-EU project employs a dual approach. On one hand, a top-down component, following the engagement stages proposed by Häberlein & Hövel (2023), which is actively led by the project partners (see Chapter 2). On the other hand, a parallel bottom-up component involving stakeholders directly expressing their interest in the FITTER-EU project through self-declaration on the project's website. This section of the Chapter details the two engagement paths of the FITTER-EU project.

Top-down engagement was initiated first, starting in April 2024, by identifying and mapping stakeholders. In parallel, coordination meetings have been held between the leaders of T2.3 and T7.2 and the project coordinator—SV, UNINOVA, and TU Dublin—to prepare the needed tools for bottom-up engagement, which will actively commence once the final version of the project's webpage is active. Both strategies will be conducted in parallel throughout the lifespan of the project, with some iterations, which are described in Subsection 4.4.2.

4.3.1 Top-down engagement path

Häberlein & Hövel (2023) propose six stages in a stakeholder engagement strategy. In the FITTER-EU project, these are part of the top-down engagement strategy being conducted in T2.3 under the guidelines provided by SV, executed in all the country-case studies led by national partners. Additionally, SV has compiled a mapping of European- level stakeholders. The main activities implemented in each of these steps are described below.

Engagement strategy steps in the FITTER-EU project

Step 1: Identification

Stakeholder identification in the FITTER-EU project revolved around three phases:

- An initial/exploratory mapping of already active contacts considering all stakeholders' types prioritised by the project in an excel sheet, shared by T2.3 leaders on April 9th (deadline April 23rd).
- A second mapping of organisations targeting structurally vulnerable groups, using an updated excel sheet prepared in coordination between T2.2 and T2.3 leaders (FGB and SV). This excel was provided by T2.2 leaders on May 3rd with a deadline of May 15th.
- A third mapping of stakeholders, requested as part of T2.3 national action plans. Guidelines developed by SV asked partners contributing to national case studies to map all stakeholder categories targeted by the project “which have a role, are interested in and/or are affected — positively or negatively— by the twin transitions and/or that operate in one or more of the four sectors prioritised by the project —building & housing, food & agriculture, transport, and energy—”. This mapping is for all stakeholders which are relevant to the project, independently of whether they are active contacts or whether their focus is on structurally vulnerable groups.

To complete the mapping of stakeholders, an integrated version of the database was prepared, unifying previous databases used for the first and second mappings. The integrated [stakeholder database](#) was uploaded to the project's SharePoint, and partners were instructed to broaden the identification in the online version to avoid creating parallel versions. Between June 18th and July 4th, six bilateral calls were conducted with all project partners leading and contributing to national case studies. These bilateral calls were dedicated to discussing stakeholder identification, active contacts relevant to the FITTER-EU project, potential challenges, and engagement incentives, as well as the use of previous database versions. These calls aimed to gather inputs to elaborate:

- The integrated version of the stakeholder database, proposing new variables and modifying existing ones. In addition to the database uploaded to SharePoint, a dictionary of variables and guidelines to fill the database were prepared and sent to partners on June 22nd, 2024, presented in Annex N° 1: Stakeholder's database.
- A template for the national action plans, included in Annex N° 2: National action plans.

Steps 2,3 and 4: Analyse, map and prioritise stakeholders

Once stakeholders have been identified, it is recommended to establish stakeholder categories to assess their potential contribution to the project (analyse step), appraise their interest and willingness to participate (mapping step), and prioritise them (prioritise step) according to their degree of potential engagement per stakeholder type (Häberlein & Hövel, 2023).

All these steps were taken by project partners while elaborating their National Action Plans. The provided template includes guidelines for appraising various dimensions per stakeholder type targeted by the project: CSO, public service/policymaking bodies (civil servants and stakeholders involved in the policymaking process), employers' organisations, business angels, private sector entities (different from employers' organisations and business angels), trade unions, and research/academic institutions. The dimensions to analyse the stakeholders are:

- Influence on policy processes related to the twin transitions (or digital and green separately, if the twin transition concept does not apply).
- Critical/anti-categorical (McCall, 2005) or agonistic approaches (Mouffe, 1999). This analytical dimension fosters the inclusion of stakeholders who might be relevant for the FITTER-EU project even if they do not formally contribute to governance structures and policy processes. For instance, this could include social actors like grass-root climate movements, or other relevant actors with no access to high-level policy making. It is worth noting that engaging these stakeholders might be challenging, as they might not be interested in working with formal policy decision-making structures. This category might be particularly relevant for CSOs, trade unions, or research/academic institutions.
- Potential contribution and relevance to the project.
- Potential interest in participating in the project.
- Key factors of participation/specific needs (determinant factors for engagement).

After analysing these dimensions in each stakeholder category targeted by the project, partners in charge of the national cases were asked to analyse their identified stakeholders in a double-entry matrix inspired by the work of Mendelow (1991). This tool requires positioning the stakeholders in a four-quadrant table according to two dimensions:

- Interest of the stakeholder in participating in the project.
- Influence these stakeholders might exert on the achievement of the project's results. Along with the 'influence on policy processes' that is typically considered, this dimension also includes relevance to the project's results and methodology – informed by a gender/intersectional and systemic approach, as this creates room for fostering intersectional inclusiveness also when prioritising and strategizing on stakeholder engagement. In fact, FITTER-EU engagement strategy aims at not being limited to reflect existing power structures and dynamics, but to enable to give marginalised voices a chance to be heard (consistent with the above-mentioned opening to stakeholders with critical/anti-categorical/agonistic approaches). Partners will prioritise the

engagement of stakeholders trying to balance between those who (already) have influence and those who should be included to achieve intersectional inclusivity.

Steps 5 and 6: Select and recruit stakeholders

Stakeholders that have been analysed and classified in the NAPs will be prioritised for each participatory activity by partners in charge of such activities or, by default, tasks leaders. Section 4.4 below is focused on the specific strategies and procedures for the recruitment of stakeholders. In addition, Chapter 6 provides specific strategies to engage stakeholders based on their position in Mendelow's matrix. Partners can refer to this chapter to proceed with the engagement of their identified stakeholder in their national action plans, having in mind the overall engagement principles outlined in Chapter 3 and with a view of applying the Inclusive Engagement Checklist (Annex N° 5: Criteria for the inclusiveness checks) as part of the continuous monitoring as explained below in section 4.5.2 of the present Chapter.

Overview of the stakeholders identified in the updated mapping of the database

Initial stakeholder mapping in the FITTER-EU project has been conducted by all FITTER-EU partners, focusing in the six countries of the consortium —Germany, Hungary, Italy, Ireland, Portugal and Spain— as well as stakeholders with the European level as geographical scope. As a result, by September 11th 2024, a total of 625 potential stakeholders to be involved in the FITTER-EU project have been identified. Stakeholders mapped belong to the stakeholder categories prioritised by the project and are focused either on issues related to the twin transition, in one of the sectors addressed by the project —energy, building & housing, food & agriculture and transport— or represent structurally vulnerable population groups at risk to the effects of either the green, or digital, or twin transition.

Most stakeholders are civil society organisations, representing 37.9% of the total (237), followed by public institutions (17.3%, with 108 mapped stakeholders). The distribution of the mapped stakeholders among the other categories is less significant, ranging from less than 4% to 11% each¹.

Mapped stakeholders are distributed in the countries of the project as follows: Germany (11.5%), Hungary (11.7%), Italy (31.4%), Ireland (19%), Spain (6.4%) and Portugal (4.8%). All of them, representing the 84.8% (530) of the database entries², have been mapped as part of the work conducted by FITTER-EU partners in the NAPs and have the national, regional or local levels as geographical scope. In particular, 475 stakeholders target the national level, while the regional and local are targeted by 28 and 20, respectively. The remaining 15.2% (95) stakeholders of the database have a focus on the European level, being located either in the countries of the project or in others such as Austria, Belgium, Switzerland, Czech Republic, Denmark, Greece, France, Lithuania, and The Netherlands.

¹ Research / academia: 68 (10.9%), Employers' organisations: 59 (9.4%), Other: 54 (8.6%), Private Sector: 42 (6.7%), Trade union: 35 (5.6%), Business angels: 22 (3.5%)

² In total, they represent 530 stakeholders, of which 196 are in Italy, 119 in Ireland, 73 in Hungary, 72 in Germany, 40 in Spain and 30 in Portugal.

Considering their work on transition, 454 (72.6%) of the mapped stakeholders focus at least on one form of transition. Slightly less than 10% (59) focus exclusively on the digital transition alone, one quarter, 160 (25.6%) is dedicated to the green transition only, whereas 235 (37.6%) work on the twin transition already.

More than one third of the mapped organisations focus on or prioritise a structurally vulnerable group (217, representing 34.7% of the total) considering, amongst others, aspects such as gender, race, poverty, migration, and age. Furthermore, 228 (36.5%) address intersectionality somehow in their mission. For example, intersectionality may be applied as a methodology to be carried out throughout the project, or by considering how different dimensions of inequality, such as gender and income, or poverty and youth, are intertwined. In addition, 367 stakeholders have been identified as participating in the policymaking structures and debates, including both actors formally participating in formal governance structures and processes, as well as actors influencing the public debate in other ways, including grassroots mobilisations.

Giving a closer look at the sectorial focus of the mapped organisations, energy has the largest share, with 227 stakeholders of the total (36.3%), followed by 173 (27.7%) in food & agriculture, 161 (25.8%) in transport, 145 (23.2%) in building & housing and 201 (32.2%) in others³. Beyond the sectors prioritised by the project, these grouped other sectors in which mapped organisations focus are, for example, artificial intelligence, waste, health and nature conservation, among others. It is worth highlighting that most of the organisations are cross-cutting: 479 (76.6%) combine a focus on the energy sector, as well as an additional sector (be it transport, building and housing, food and agriculture or the category “other”). It is also worth highlighting that some of them cover more than two sectors.

The following table zooms in the prioritised sectors, analysing in each one of them, the distribution of stakeholder categories.

³ It is worth clarifying that focus on the project sectors have been included in the database as dummy variables, allowing to report the contemporary focus of one organisation in more than one sector. It is important to note also that many of these variables have been left as blank in the mapped stakeholders. For the calculations, blanks have considered as “no”; therefore, percentages are referred to the entries in the dataset which have been filled with “yes” in their sectoral focus.

Table 8: Sectoral distribution of stakeholder types

Type of organisation	Total organisations		Focus on energy sector		Focus on transport Sector		Focus on building & housing sector		Focus on food & agriculture sector	
	N°	%	N°	%	N°	%	N°	%	N°	%
Business angel	22	3.5%	8	4%	6	3.7%	3	2.1%	8	4.6%
Civil society organisation	237	37.9%	53	23%	22	13.7%	31	21.4%	38	22%
Employers' organisation	59	9.4%	22	10%	19	11.8%	17	11.7%	22	12.7%
Other	54	8.6%	17	7%	15	9.3%	11	7.6%	19	11%
Private sector	42	6.7%	20	9%	9	5.6%	11	7.6%	7	4.1%
Public service / policymaking	108	17.3%	54	24%	47	29.2%	43	29.7%	46	26.6%
Research/academia	68	10.9%	37	16%	25	15.5%	17	11.7%	21	12.1%
Trade union	35	5.6%	16	7%	18	11.2%	12	8.3%	12	6.9%
Total	625	100%	227	100%	161	100%	145	100%	173	100%

As stated above, when analysing all the information included in the database, civil society organisations are the most widespread type of stakeholder mapped, followed by actors belonging to the category public service/policymaking. The distribution of stakeholders in each category changes when considering the focus on specific sectors. In particular, among organisations with a sectoral focus, policymakers/public service is the most common type of stakeholder with 190 of the 709 stakeholders with a sectoral focus (26.9%)⁴. Likewise, in all the sectors, civil society organisations are the second most common type among the stakeholders that have a sectoral focus. This divergence in the distributions between the organisations with a sectoral focus and the total of the mapped stakeholders might be due to the fact that public institutions are more easily identifiable with a specific sector, because their mission is usually specific.

With the objective to take a closer look at the country differences, the following a table presents the distribution of stakeholder types per country.

⁴ Considering specific sectors, this percentage is of 24% in energy, 27% in food & agriculture, 29% in transport and 30% in building & housing.

Table 9: National distribution of stakeholder types

Type of organisation	Germany		Hungary		Ireland		Italy		Portugal		Spain		European level organisations	
	N°	%	N°	%	N°	%	N°	%	N°	%	N°	%	N°	%
Business angel	2	2.8%	3	4.1%	1	0.8%	8	4.1%	0	/	1	2.5%	7	7.4%
Civil society organisation	34	47.2%	25	34.2%	76	63.9%	35	17.9%	13	43.3%	11	27.5%	43	45.3%
Employers' organisation	4	5.6%	5	6.8%	6	5%	39	19.9%	0	/	4	10%	1	1.1%
Other	6	8.3%	0	/	3	2.5%	19	9.7%	10	33.3%	2	5%	14	14.7%
Private sector	7	9.7%	4	5.5%	4	3.4%	13	6.6%	2	6.7%	9	22.5%	3	3.2%
Public service / policymaking	10	13.9%	4	5.5%	20	16.8%	48	24.5%	4	13.3%	10	25%	12	12.6%
Research/academia	4	5.6%	19	26%	7	5.9%	24	12.2%	0	/	1	2.5%	13	13.7%
Trade union	5	6.9%	13	17.8%	2	1.7%	10	5.1%	1	3.3%	2	5%	2	2.1%
Total	72	100%	73	100%	119	100%	196	100%	30	100%	40	100%	95	100%

In all countries except for Italy, where public service and / policymaking stakeholders prevail, the most diffused category of mapped stakeholders are CSOs. This category is particularly relevant for Ireland (64%), Germany (47%) and Portugal (43%). The other three countries have their mapped stakeholders more evenly distributed among other stakeholder categories beyond CSOs: research/academia and trade unions for Hungary (26% and 18%), public service/policymaking and employer's organisations for Italy (25% and 20%) and public service/policymaking and private sector for Spain (25% and 23%).

With regard to the stakeholders with the European level as geographical scope, of the 95 stakeholders mapped, the vast majority are civil society organisations: 43 (45.3%). Fourteen fall under the umbrella category "other", whereby their share is relatively high, because they are complex networks that cannot be clearly and exclusively characterised as a specific type of stakeholder – exemplary the *EEAC, The European Environment and Sustainable Development Advisory Councils Network*, which is defined as a 'knowledge community'. The next highest share consists of thirteen stakeholders belonging to the research or academic typology.

Considering their sectoral focus, 65 (68.4%) are from the energy sector, 30 (31.6%) from transport, 24 (25.3%) from building & housing, 42 (44.2%) from food & agriculture. In addition, due to their cross-cutting nature, almost two thirds —63 (66.3%)— focus on sectors other than the ones prioritised by the project.

4.3.2 Bottom-up engagement path

The bottom-up path of the FITTER-EU engagement strategy is based on stakeholders' self-declaration of interest to participate in the CCGS via a registration form on the project's website. As the website has not been yet finalised, this path is currently inactive and will be activated once the final version is released. Despite this, since April 2024, SV has been coordinating this activity with UNINOVA, the partner responsible for communication and T7.2 leader, who manages the CCGS.

SV has prepared and delivered the following set of materials to UNINOVA on June 28th, 2024, to be integrated into the webpage's "collaboration" section:

- Invitation text to join the CCGS to be included in the project's website.
- Infographic on how to become part of the CCGS.

- An informed consent form to be enclosed to the registration form
- Registration form: self-declaration of interest to become part of the CCGS and collaborate with the FITTER-EU project, interoperable with the stakeholders' database.

All the aforementioned documents are present in Annex N°3. As far as the registration form is concerned, all its entries correspond to the variables included in the database (see Annex N° 1: Stakeholder's database for the dictionary of variables of the database), with the exception of those dedicated to monitoring the engagement (variables AJ to AO) and variables to be filled by partners about the stakeholder's engagement processes in the project (AE and AF) and their contact status (AG, AH and AI). The registration form is included in the "collaboration" section of the website.

By filling out the registration form, a person confirms their interest in collaborating with the FITTER-EU project, meaning their contact details will be added to the stakeholder database. Consequently, they may be contacted to participate in the project's participatory activities if the partners in charge consider their expertise relevant and based on the needs of the different tasks. Additionally, as specified in the informed consent form, the information included in the stakeholder database will not be shared outside the FITTER-EU consortium, and will be stored on the project's SharePoint, which is GDPR compliant.

In Section 4.4, the specific procedures and processes to be followed by partners in tasks where stakeholder engagement is foreseen, are explained more in detail.

4.4 Procedures for the engagement of stakeholders

This Section describes the specific procedures and provides guidelines for the recruitment stage of the stakeholder engagement strategy of the FITTER-EU project.

4.4.1 When to start the FITTER-EU stakeholder engagement

Top-down stakeholder engagement

The top-down stakeholder engagement strategy is closely linked to the development of the participatory tasks of the FITTER-EU methodology. The times when the project's engagement efforts are planned are outlined in the calendar of participatory activities in Subsection 4.2.2. However, as already noted, in most cases the dates indicated by partners are planned as ranges of more than one month, since the participatory tasks of the methodology only started in September 2024.

It is recommended to start active stakeholder engagement, i.e. **establishing contact with potential participants in the participatory tasks of the FITTER-EU methodology, at least two months before the activity takes place**. Task leaders and T2.3 leader (SV) will coordinate in advance when tasks involve stakeholder engagement and give specific instructions on how to proceed in terms of timing and invitation formats. T7.2 leaders (UNINOVA) will also be involved when needed for related communication activities via the official FITTER-EU dissemination channels. **Partners in charge of national cases should wait for these explicit instructions before contacting specific stakeholders and involving them in a task**. Subsection 4.4.2 below describes the specific procedures all task leaders should follow when a task involves stakeholder engagement.

Bottom-up stakeholder engagement

The bottom-up path of the FITTER-EU engagement strategy runs parallel to the top-down path. Unlike the top-down approach, which focuses on involving stakeholders for the project's specific tasks, the bottom-

up approach aims to maximise the project's reach, communicating with a broad number of stakeholders who might be interested in being updated and/or participating into the CCGS.

A prerequisite for bottom-up engagement is the launch of the final version of the project's website with its "collaboration" section, where stakeholders will be able to fill out a form self-declaring their interest in participating in the project.

This path of engagement should start after September 2024, once partners have access to the final version of the stakeholder engagement guidelines in D2.5, and as soon as the final version of the website is launched online.

Throughout the project, bottom-up and top-down engagement paths will iterate. To facilitate this, it is recommended to implement "inclusiveness checks" (see Section 4.5.2) every six months to identify possible gaps in stakeholder engagement. After each inclusiveness check, it is recommended to conduct a new bottom-up engagement round, supported by communication/dissemination efforts at the project level in alignment with WP7 leaders (i.e. through dedicated news items and/or blog posts). A tentative schedule is provided below:

- M9 (October 2024): Start of the bottom-up engagement.
- M15 (April 2024): inclusiveness check and second iteration of bottom-up engagement.
- M21 (October 2025): inclusiveness check and third iteration of bottom-up engagement.
- M27 (April 2026): inclusiveness check and fourth iteration of bottom-up engagement.
- M33 (October 2026): inclusiveness check and fifth iteration of bottom-up engagement.

For a smooth process and for maximising the integrated top-down/bottom-up approach, the entries/contacts collected via the website form for the bottom-up path will be integrated into the Stakeholder Database on a regular basis (see Step 10 in Table 10).

4.4.2 Process to follow when a task foresees stakeholder engagement

The following table details the procedure to be followed by task leaders in each task that foresees stakeholder engagement.

Table 10: FITTER-EU top -down internal procedures

N°	Steps/ milestones	Timing	Activities	Partners involved	Materials to be prepared and available templates
1	Start of top-down engagement	At least 2 months before the event	Initial communication via e-mail from task leaders to SV about the specific timeline foreseen for the participatory activity and to coordinate on step 3	Task leader that foresees the participatory activities SV	
2	Drafting the content/specific methodology and tools of the participatory activity	From 3 to 2.5 months before the event	Task leaders of the participatory activity draft a rich presentation of the event/task outlining in a clear way: objectives of the activity, structure of the session, speakers/facilitators, what specific contribution is expected from participants and the estimated time to prepare including all preliminary informative materials.	Task leader that foresees the participatory activities	Presentation of the event (PPT) or outline of the session (template) Available templates: Annex N° 4: Templates for stakeholder engagement Invitation letter, informed consent form, registration form (G form), ex post feedback survey (G form) and follow-up email.
3	Meeting between task leaders and SV	Approximately one week after the materials are sent to SV	A dedicated meeting to discuss the task specific engagement strategy, including messages/incentives. Agenda and content of the event discussed if relevant.	SV Task leaders	
4	Review of documents	Maximum 1 week after the event	Review of the materials prepared for the participatory activity by SV and UNINOVA. Suggestions sent to task leaders for the final validation	SV UNINOVA Task leaders	

N°	Steps/ milestones	Timing	Activities	Partners involved	Materials to be prepared and available templates
5	Translation of materials to national languages	Two months before the event	Partners in charge of National Case Studies translate materials	Partners in charge of national cases	Reviewed and translated materials
6	Communication/ dissemination strategy (only if the event is open, otherwise skip to step 8: invitation through direct contact)	Two months before the event	UNINOVA suggests a communication/ dissemination strategy and related channels. In parallel, UNINOVA publishes events on the website with a related form (possible use of G Forms) where participants can register. Registration forms are shared automatically with partners.	UNINOVA National partners (in charge of customising the registration form)	News' item on the event on the FITTER-EU website Registration form
7	Dissemination of the event (only if the event is open, otherwise skip to step 8: invitation through direct contact)	Starting two months before the event	Create news items and/or social media posts. Follow other guidelines provided by UNINOVA in conjunction with TFC in the communication/ diffusion strategy	All partners	News items and/or social media posts
8	Direct contact with selected stakeholders	Starting two months before the event	Each partner in charge of the national cases makes use of the online version of the stakeholder's database and filter their national contacts to select a pool of potential contacts to be involved in the specific activity. Filters are used according to the guidelines provided by SV and task leaders.	Partners in charge of national cases. In case there is more than one partner in the national case: see coordination	

N°	Steps/ milestones	Timing	Activities	Partners involved	Materials to be prepared and available templates
			When sending the invitation, link to the webpage and the registration form are included and a defined time is given to confirm the attendance of maximum 1.5 weeks. After, a reminder is sent to the contacted stakeholders.	mechanisms in Subsection 4.4.3	
9	Interim monitoring of the registration	3 weeks before the event	Analysis of the registrations and strengthened communication efforts if needed, or refined communication messages based on received feedback	SV UNINOVA Task leaders/national partners	
10	Update of the stakeholder's database (see Section 4.5)	No longer than a week after the event	Update of the database integrating the new stakeholders who participated and filling the columns dedicated to monitoring	Partners in charge of national cases Task leader	

Bottom-up stakeholder engagement

As mentioned in the previous section, it is recommended to start the bottom-up stakeholder engagement path of the FITTER-EU project after September — once partners will receive complete guidelines with D2.5— as soon as the FITTER-EU webpage is launched in its final version.

Table 11: FITTER-EU bottom-up internal procedures

N°	Steps/ milestones	Timing	Activities	Partners involved	Materials to be prepared and available templates
1	Selection of contacts for the bottom-up engagement	September 2024, as soon as the final version of the project's webpage is launched online.	Select a pool of contacts from the stakeholder's database to be contacted and invited to collaborate with the FITTER-EU project	Partners in charge of national cases in coordination with the other partners from their countries (see Subsection 4.4.3)	Annex N° 3: Initial developments to operationalise the bottom-up stakeholder engagement path

N°	Steps/ milestones	Timing	Activities	Partners involved	Materials to be prepared and available templates
		SV in coordination with UNINOVA will provide the specific indication on when to start.		Partners with direct contacts	• Invitation letter to become part of the CCGS. • Informed consent form
2	Translate the invitation letter	Every time during the iteration (see Subsection 0)	Translate the invitation letter provided in Annex N° 3. Stakeholders should be invited in their language.	Partners in charge of national cases	
5	Blogpost or news published in the project's webpage and social media		Produce communication materials for the launching of the bottom-up stakeholder engagement. SV will produce the text for a blogpost, while UNINOVA the illustrations and social media posts.	SV UNINOVA	Communication materials
4	Direct contact with selected stakeholders		Send via email the invitation letter, the link to the webpage's "collaboration" section and the registration link. Give a defined time to confirm the attendance of maximum 1.5 – 2 weeks. After this time send a reminder to the contacted stakeholders. Check to identify the ones that have replied or not to your invitation	Partners in charge of national cases Partners with direct contacts	
5	Update of the database	3-4 weeks after partners start with the bottom-up engagement.	UNINOVA will provide access to the database created with the registration form's replies since the beginning of the process. The registration form will contain the same variables than the integrated	Partners in charge of national cases, SV for EU stakeholders	

N°	Steps/ milestones	Timing	Activities	Partners involved	Materials to be prepared and available templates
		Periodically every 3 months since the bottom-up stakeholder engagement starts.	stakeholder's database that is being co-constructed as part of the top-down engagement; therefore, the integration should be easy.		

The bottom-up engagement process needs a permanent and continuous coordination between T2.3 and T7.2 leaders to be successful. The next steps below are considered prerequisites for implementing the bottom-up engagement strategy:

- Finalise the online versions of the webpage and the “collaboration” section with the registration form, in order to start with the bottom-up stakeholder engagement.
- Give access to the replies to the online registration form to all partners since the launching of the “collaboration” section.
- Update the stakeholder’s database with the contacts from the bottom-up engagement every three months since the start of the bottom-up engagement.
- Define the process for updating the database between UNINOVA and Smart Venice.
- If technically feasible, it is suggested to implement a CRM (Customer Relation Management) to manage more efficiently the database of contacts and the integration of the two databases.

4.4.3 Coordination process for engagement activities in national cases with more than one partner

The FITTER-EU project has six national cases, each with a partner in charge of coordinating the related research activities. However, there are three countries where more than one partner contributes:

- Germany (WECF)
- Hungary (SFC)
- Italy (FGB), contribution of SV.
- Ireland (TU Dublin), contribution of TFC.
- Portugal (UNINOVA)
- Spain (UCM), contribution of AITEC and Eurecat.

In cases where there is more than one national partner, it is fundamental to coordinate the engagement efforts for activities such as stakeholder mapping and the elaboration of the national action plans, translating communication materials, contacting stakeholders, and updating the database.

It is recommended that the partner in charge of the country case calls for an internal coordination meeting with the other national partners at the beginning of the engagement process - once the indication to start the engagement efforts for a specific project activity is given by the task leader (in the top-down engagement path) or by SV (in the bottom-up engagement path). Some recommendations apply:

- Each partner engages its personal/direct contacts.
- Engagement of different types of actors might be split between the partners.
- Leading the engagement might be split between partners per task (considering the task leadership).

Different arrangements are also possible; what is important is to ensure coordination to avoid duplication of efforts or contacting the same stakeholder multiple times.

4.4.4 General recommendations for the preparation of tasks with participatory components

In what follows, some general recommendations to plan participatory activities are presented. Consider that materials for the engagement of participatory tasks will be prepared by task leaders —see Annex N° 4: Templates for stakeholder engagement—. The steps to be followed to implement the stakeholder engagement, describing the task distribution between task leaders and partners in charge of country cases are detailed in Table 10.

- **Invitations and pre-event information:**
 - When sending invitations, include a clear and concise agenda along with additional information about the objectives, the methods in use, and expected content of the discussion: it is important to provide participants with clarity on what they will get from participating, and which are the expectations towards them.
 - Send a brief preparatory document (1-3 pages) to ensure stakeholders are well-prepared. This will improve the quality of the inputs collected during the session and enrich the discussions. Avoid sending long documents, which might discourage stakeholders reading them.
 - Attach the Informed consent form (see Annex N° 4: Templates for stakeholder engagement) to make the terms and conditions of participation and personal data treatment conditions clear.
 - Include visuals or infographics to improve clarity of the information and make the information more appealing.
 - Refer to who else has been invited: for group activities, knowing whom they will have the chance to meet with can be a strong networking incentive.
 - Make a registration form for the event where you ask about inclusiveness needs, such as accessibility requirements.
- **Agenda planning**
 - Allocate sufficient time for the activities: at least 10-15 minutes for groups discussions both for in-presence and online (breakout rooms) formats.
 - Alternate between group work and plenary sessions to deepen discussions and ensure inclusivity.
 - If moments for silent reflection are foreseen in the agenda of the event, avoid combining them with moments for coffee breaks, as this could lead to distractions.
- **Session dynamics**
 - At the start of the session, it might be worth sharing some “house-rules” establishing some guidelines to conduct the discussion and session dynamics, ensuring inclusiveness and respectful discussions among participants. It is also recommended to ask for the personal pronouns of participants.
- **Tracking and proving participation, communication related matters**
 - Signatures of participants will have to be collected for in presence events while for online events, screenshots of the participants list on the platform in use should be taken. It is recommended to save these documents/files on partners’ own repositories/archives in case of Technical Audits from the EC.
 - If the event will be disseminated via social media/websites posts or news it is important to integrate the Informed consent form a specific option to participants to express/deny consent to the use of pictures and to pay attention when photo shooting/selecting imaged that the will of participants is respected.

4.4.5 Engagement of European stakeholders, Sister projects and role of the FITTER-EU External Advisory Board

The FITTER-EU project methodology is mostly centred in studying national case studies in the six countries covered by the project. Still, as twin transition policies are mostly framed within EU level policy agendas (e.g. the European Green Deal, the Europe Digital Decade Policy Programme, the Resilience and Recovery mechanism and NextGeneration EU funds, Horizon Europe research agendas) the framing and State of the

Art research performed in WP2 has been taking into account EU policies as a context for the national case studies. Along these lines and even if a proper EU level case study falls out of the FITTER-EU's remit, the Stakeholders Database has been developed with a dedicated section mapping contacts of European level stakeholders. EU level stakeholders can be identified in the database with the label "European" in variable G "Geographical scope".

These stakeholders could play an important role in the overall engagement strategy of the project:

- They could be consulted by partners on ad-hoc basis in the course of the delivery of the research and innovation components of the project to get expert input and advice.
- They could be intermediaries to reach out to national level stakeholders/disadvantaged groups (DGs): this could be the case of membership organisations/EU level CSOs in particular.
- Their presence in the frame of participatory events in WP3, 4, 5 and 6, and/or connected dissemination events in WP7, has the potential to function as an incentive for the participation of national level stakeholders who might have the interest to learn about policy/advocacy developments and networking or funding opportunities.

Top-down engagement of EU level stakeholder has already started through the mapping exercise initiated by SV via the database and will be planned in detail in the course of the project, based on the different Tasks' needs and in agreement with WP and Task Leaders. Update of the Stakeholders database for EU level Stakeholders will be taken care of by individual task leaders including tracking those stakeholders who become part of Expert Group as they have participated to specific events/project's activities. Contact and invitations of European stakeholders to participatory activities of the project will be conducted by task leaders following the procedures established in Table 10.

A bottom-up engagement path will also be delivered following the steps foreseen in Table 11 for this type of stakeholders, taken care of by SV, with TUD to be included in cc to communications/invitations.

Sister project engagements will be led by TFC in conjunction with TU-Dublin. It is envisaged that the question of how to better collaborate as a unit will be ascertained in joint workshops. Sister projects clustering is based on the principle of sharing common knowledge and understandings so that as a unit, FITTER and the sister projects are not working in silo mode. Currently there are two sister projects ST4TE and READJUST, both complementary to FITTER-EU. [ST4TE](#) is led by Aristotle University of Thessaloniki (Greece). It aims to provide a comprehensive view of the drivers of the twin transition (TT), the inequalities that emerge or are widened by it, and a set of policies to build greener, more equal and more productive societies. [READJUST](#) is led by YAGHMA B.V. (The Netherlands). It aims to track, assess, analyse and address a wide range of aspects of intended and unintended inequalities triggered by twin transitions policies.

FITTER-EU has set up an External Advisory Board (EAB). Engagement is overseen by TFC in conjunction with TU-Dublin. The board is composed of 7 high-profile experts from research, policy, and practice domains that are relevant to the project's scope. Three of them are from institutions based in partners' countries (HU, PT, DE), one of them works at a European level research institution, one at a research organisation from UK with a global scope. These contacts are included in the Stakeholders Database as well as on the FITTER-EU web site. The seven members of the Board are:

1. Dr. Becky Faith who is a Research Fellow in the Digital and Technology Research Cluster at the Institute of Development Studies at the University of Sussex.
2. Zsuzsanna Koritár, who has been working in the field of sustainable energy management in the non-governmental sector since 2001.

3. Gonzalo Génova who is Associate Professor at the Computer Science and Engineering Department of the Carlos III University of Madrid.
4. Dr. Irina Velicu who is a political scientist working on socio-environmental conflicts in Europe, particularly post-socialist countries, environmental justice, social transformation and aesthetic politics.
5. Alessandro Giordano who is a Scientific Officer at the European Commission's Joint Research Centre (JRC) in Seville, Spain.
6. Prof. Dr. Bernhard Kölmel who has held the Chair of Global Process Management at the University of Pforzheim, Faculty of Engineering, since 2012.
7. Ion Dogeanu is a Top C-Level 20+ manager, from 2007 acting as Sustainable development evangelist from the leading position of the Energy Agency in Bucharest Romania.

Engagement with the members of Board will be conducted in both virtual and in person mode and the undertaking will be held at key milestones. Not all members will necessarily be invited to participate at each milestone. This is dependent on their individual background and the primary subject matter of the work conducted in the FITTER project. Feedback from the EAB meetings will be recorded in a report prepared by TFC and discussed at the monthly project review meetings that are chaired by TU Dublin.

4.5 Monitoring

The present Section is dedicated to the monitoring of stakeholders engagement a component that is emerging as crucial to increase quality, meaningfulness, reflexivity and inclusiveness of methods and strategies in the field, as highlighted by existing research and best practices summarised in Chapters 2 and 3. Monitoring consists of four components, each detailed in the subsections below: update of the database, periodic internal inclusiveness checks, the collection of participant's feedback after the participatory activities and the monitoring of the quality of the received inputs in the participatory activities.

4.5.1 Stakeholder's database update after participatory activities (task ownership: each partner in charge of national case studies, coordination by UNINOVA)

As described in Subsection 4.4.2, the stakeholder's database should be updated after establishing the contact with stakeholders:

In top-down stakeholder engagement:

- The database will be updated by partners in charge of the national cases and tasks leaders (the latter only in the case when they have contacted stakeholders directly). This should occur within the week following the participatory activity.
- The update of the database implies new variables dedicated to monitoring of the engagement (clarified in a dedicated "guidelines sub-sheet) and the rationale is to track whom of the invited stakeholders actually took part as well as adding new stakeholders in case they registered and joined the event even if not invited. These should be filled in once an existing stakeholder is contacted and participates in a project activity. Additionally, new stakeholders should be included in the database if they had not been added previously. This way, the Database works in a similar way as a CRM tool and can support tracking engagement and calculating actual participation metrics for the technical reporting.
- In particular, the variables to be modified in the database are:
 - Column "AJ": Already contacted for the project. Indicate yes in the ones contacted for the participatory activity.

- Column “AK”: Declared interest in participating. Indicate yes/no according to the answer of the contacted stakeholder.
- Column “AM”: Task participation. For the contacted stakeholders that accepted to participate include the participatory activity “T3.1” for example.

In bottom-up stakeholder engagement:

- As detailed in Subsection 4.4.2, the recommended periodicity for the bottom-up stakeholder engagement should be every six months. Invitations to new stakeholders should be sent with input from inclusiveness checks according to the following proposed calendar:
 - M9 (October 2024): Start of the bottom-up engagement.
 - M15 (April 2024): inclusiveness check and second iteration of bottom-up engagement.
 - M21 (October 2025): inclusiveness check and third iteration of bottom-up engagement.
 - M27 (April 2026): inclusiveness check and fourth iteration of bottom-up engagement.
 - M33 (October 2026): inclusiveness check and fifth iteration of bottom-up engagement.
- To assure the interoperability between the information collected in the bottom-up registration form—which will be available in the “collaboration” section of the FITTER-EU webpage—, all questions included in the former will directly reflect variables of the latter. This would facilitate merging the two databases.
- Update of the stakeholder’s database with the new registrants in the FITTER-EU webpage will be managed centrally by SV in coordination with UNINOVA.

4.5.2 Periodic internal inclusiveness checks (task ownership: each partner in charge of national case studies, coordination by SV and TFC)

To assure the inclusiveness and representativeness of the engaged stakeholders both at the level of the stakeholder’s database and the actually engaged stakeholders, periodic inclusiveness checks will be implemented as part of the monitoring of the FITTER-EU stakeholder engagement strategy every four months, following the workflow below:

Partners answer to the inclusiveness checklist in Annex N° 5: Criteria for the inclusiveness checks and send results to SV.

Smart Venice appraises data from the stakeholders database, considering the distribution in terms of sectors, representation of structurally vulnerable groups and types of organisations, encompassing the stakeholders that have been mapped and the ones effectively involved in the FITTER-EU project.

The recommended periodicity for this is every 6 months:

- M10 (November 2024): first inclusiveness check (one month after the first iteration of bottom-up stakeholder engagement).
- M15 (April 2024): second inclusiveness check.
- M21 (October 2025): third inclusiveness check.
- M27 (April 2026): fourth inclusiveness check.
- M33 (October 2026): fifth inclusiveness check.

After the input received from SV, partners in charge of national cases could orient their efforts to involve categories of stakeholders in which there might be some gaps. If mapped stakeholders have been contacted and declined participation, national case studies partners will reflect on possible causes and

improvements in their engagement strategies and expand the stakeholder list to include additional contacts to help fill the gaps.

4.5.3 Participants' feedback collection and accountability towards engaged stakeholders (task ownership: each partner in charge of national case studies, coordination by UNINOVA)

This monitoring action is extremely important for all group activities from Clusters 2 to 5, it stems as a recommendation from literature and best practices on stakeholder engagement and it will focus on collecting feedback from participants about how they have perceived their experience with FITTER-EU. This will enable ensuring continuous improvement in the implementation of the engagement strategy and it is linked to an approach geared towards being accountable towards stakeholders who have entrusted us dedicating their pro-bono time and expertise/knowledge. UNINOVA will be in charge of steering this component of the monitoring process.

The workflow to operationalise this would function as following:

- The survey template draft in Annex N° 4: Templates for stakeholder engagement (available at this [Link](#)) is updated/adapted by UNINOVA and Task Leaders jointly with details on the specific event. Each national partner/team can opt in for translations in national languages/further adaptations, as needed.
- As the event is about to close, the survey is shared with participants via a QR code/link and participants are invited to answer on the spot, before they leave the venue/online platform (this should maximise chances they reply).
- One day maximum after the event, each partner in charge of national case studies sends a follow-up email to all participants to thank them for their participation, sharing key insights from the event if it was a collective one, enclosing materials if relevant/available (slide-sets, etc.), recapping on how results from the event will be used in the FITTER-EU R&I plan and by when/how they are expected to be made available. Such a follow up message can also include a reminder with link to fill the online survey for those who didn't do it yet.
- Results from the surveys will be consulted by partners individually and key highlights shared within Project Meetings to enhance the quality of the engagement strategy on an ongoing basis. Results will not be subject to comprehensive analysis as this would be beyond the scope of the project.

4.5.4 Monitoring the quality of the received input by stakeholders in the frame of the FITTER-EU R&I actions (ownership by Task Leaders in WP3-4-5-6, coordination by TUD and TFC)

This task, as obvious as it might seem, is crucial to assess to what point the engagement efforts lead to high quality input from stakeholders. Based on constant feedback/input from all partners, Task Leaders will cross-check on this aspect and reflect/make suggestions on whether/how the engagement strategies could be adjusted and refined in the different countries. This step will follow the individual Tasks' timelines and the above-mentioned reflections/suggestions can be shared either in PM meetings or in WP's leaders or meetings.

5 Challenges and incentives for engagement

5.1 Challenges documented in the literature

The complexity of engaging stakeholders in the twin transition poses various challenges. Forsythe et al. (2016) outline the challenges and risks of stakeholder engagement. The five most reported by their sample (in order by most commonly reported) are lack of stakeholder time, lack of researcher time, lack of stakeholder training/background, difficulty finding appropriate representatives to engage, and lack of research team resources. The mitigation strategies suggested include “allowing ample opportunity for stakeholders to ask questions, communicating through multiple channels, having face-to-face meetings, making time to socialise, and training researchers on engagement” (p. 16). In this chapter, potential challenges in stakeholder engagement are presented, alongside mitigation strategies and incentives.

5.1.1 Challenges related to an unsystematic engagement strategy

One of the primary challenges in stakeholder engagement is the need for a clear definition of stakeholder engagement itself. Kujala et al. (2022) note that “the lack of a shared understanding of the essentials of stakeholder engagement and its variance to related constructs hinders the progress of stakeholder engagement research” (p. 1138).

The definition is a prerequisite for effective co-creation and subsequent evaluation of the process, ranging from the terminology to scope and impact. Braun & Starkbaum (2023) note that “in most of the Responsible Research and Innovation discourse, ‘stakeholders’ are used as a catchall phrase denoting societal actors without actually pointing to who they are, why their participation is important, what exactly they contribute and why they should be involved in the [...] processes” (p. 230). Martinez et al. (2019b) describe “challenges to measuring stakeholder engagement [as] the lack of: a unifying conceptual model of stakeholder-engaged research; consensus on how to define meaningful engagement; a common understanding of who stakeholders are and what groups they may represent and unifying definition of what constitutes an engaged stakeholder” (p. 1337).

To address such hurdles, Chapters 2, 3 and 4 in this deliverable have focused on framing the theoretical background and defining concepts and principles as well as operationalising them by tailoring for the FITTER-EU research methodology.

Another aspect of unsystematically approaching stakeholder engagement is found to be the lack of evaluation: in fact, final evaluation and its reviews as well as consistent evaluation during the research, are found to function as a vehicle of improvement. Evaluation should be designed and conducted with reference to the participants, goals, mechanisms and timeframes of the broader engagement process. Ideally, “measures should be taken at the beginning of engagement, to allow a baseline for comparison of later measurements” (Gardner et al., 2009, p. 13). Providing regular updates on progress and celebrating successes can maintain momentum and encourage continuous improvement. As Hering et al. (2012) suggest, “a key concern throughout all assessment processes [is the] inherent tendency of individuals and institutions to respond to indicators as incentives. That is, the indicator itself becomes the goal rather than only corresponding to it. This has the potential that effort will be directed toward ‘gaming the system’ rather than solving actual problems” (p. 97). In view of this challenge, we have suggested monitoring and evaluation processes that do not over-burden researchers.

5.1.2 Emerging conflicts

Given the diversity of stakeholders involved, with different interests and visions, conflict is likely to occur. However, conflict can be managed by adopting a flexible and adaptive approach, allowing for compromise, iteration and facilitating open dialogues where stakeholders can express their concerns and suggestions. Mediation and conflict resolution mechanisms should be in place to address disagreements constructively. Working with diverse stakeholders across Europe also includes differing native languages, values, objectives, and interests, which necessitates a high degree of cultural competence and adaptability. To bridge gaps, it is beneficial to leverage on cross-cultural skills of facilitators and encourage mutual exchange.

Ethical concerns arise from the fact that the building of trust, commitment, and power differentials may put researchers and stakeholders on different planes: “engagement is intensive in time, resource and skill requirements, and engagement involves giving up a degree of control to people beyond the instigating group or organisation, which can threaten the adoption of a preferred outcome” (Gardner et al., 2009, p. 12).

5.1.3 Discontinuity and inconsistency

Despite high levels of engagement being time consuming, there is a need for the engagement approach to be tailored to the project and customised to the respective stakeholders. “Infrequency of engagement [...] is discouraging [and] can lead to unwarranted challenges: the research can result in policy and other decisions that are ill-fitted to the interests, opinions, and needs of [the stakeholders], and can result in fierce opposition that derails well-intentioned decision-making” (Concannon et al., 2014, p. 1697).

Establishing trust is a vital challenge, especially when discussing topics related to the twin transition, because “when stakeholders are engaged in discussions about climate change, they can become alarmed, fearful, and concerned about their future before moving onto identifying possible solutions. Such findings highlight the importance of engaging stakeholders in an appropriate manner to ensure that interactions are positive and enabling rather than the opposite” (Gardner et al., 2009, p. 8).

Countering uncertainty about the disadvantages resulting from the twin transition requires a **robust strategy to promote scientific literacy and critical thinking**. **Effective communication** requires messages to be tailored to a diverse audience, using clear and relatable language, and leveraging multiple communication platforms. Visual aids, storytelling, and highlighting local impacts can make communication strategies more engaging and accessible, as resistance to change is a natural human tendency.

To counter resistances, change management strategies, such as engaging stakeholders early in the process, providing clear rationales for changes, and supporting a variety of stakeholders continuously, are crucial. For this reason, the inherent feature of flexibility is essential for researchers, among others, to account for power hierarchies, and enable an iterative engagement process. The challenge of not reproducing hierarchies between stakeholders is critical for genuine collaboration. This can be mitigated by creating egalitarian structures, such as rotating leadership roles, ensuring diverse representation in decision-making bodies, and fostering an inclusive culture where everyone feels valued and heard. As a result, a useful approach is the **conflict resolution approach**, recognising that there are power imbalances, disagreements and conflict, and that it is important to uncover underlying fears and assumptions to collaborate effectively. For example, D2.1 outlined that “From an energy justice perspective misrecognition has been shown to occur when local opposition to environmental policies is attributed to ignorance and/or self-interest by policy makers with insufficient evidence base for such conclusions. Taking the time to understand the core concerns of those who oppose environmental policies, developing a broad based understanding of the

assumptions, values and judgements that underpin distinct positions in relation to a policy (in favour, against, undecided) and taking a conflict resolution approach to deliberative public engagement could arguably lead to the creation of a negotiated compromise that does not sacrifice social justice requirements in order to achieve environmental and economic goals (Barry et al, 2008)” (p. 77).

An important feature of the FITTER project is the diverse involvement of Civil Society Organisations (CSOs) and citizens. Their absence in stakeholder engagement would hold the risk of open conflicts, as these groups often represent the interests and concerns of the broader public. Therefore, it is essential to encourage inclusive participation by ensuring that all voices are heard from the beginning. This can be achieved through public consultations, participatory workshops, and the creation of the FITTER platform for dialogue, to contribute meaningfully to inclusive decision-making processes in the twin transition. Stakeholder engagement calls for trust and commitment among researchers and stakeholders, to overcome ethical concerns, stakeholders’ distress, as well as cultural differences (Concannon et al., 2014).

5.1.4 Flawed communication and technical barriers

Concannon et al. (2014) explain that the “inclusion of a broad range of stakeholders in research is not without its challenges. These may range from logistical (e.g., scheduling and funding) to process-oriented (e.g., balancing competing interests, addressing implicit power differentials, and managing conflict). Indeed, the main reported barriers to engaging stakeholders [...] were time and the need for logistical support and funding flexibility” (p. 1698). When it comes to engaging with stakeholders, “one of the biggest challenges [...] is reaching out and involving stakeholders who are not usually involved in such projects, or are less yet convinced by the issue, or the project/activity proposed” (TeRRIFICA, 2019, p. 26). Additionally, different stakeholders may hold competing priorities, such as, among others, varied timescales (Hinchcliff et al., 2014), or diverging/conflicting interests or socio-political frames. For example, in discussions concerning sustainability and related policy choices, the tension between public and economic interests often arises, framed as ‘money versus environment’. This dichotomy can be mitigated by leveraging on the systemic and complexity-oriented approach that is featuring the FITTER-EU methodology (see D2.1). Facilitators within group discussions should be prepared to let diverging interpretations emerge. In this way, both shortcomings and potential benefits/impacts beyond the short term can be addressed, as well as the interdependencies among the different dimensions and variables under scrutiny. Emphasis on dialogue and managing it in an inclusive and balanced way is important: this can be achieved by stressing that grasping divergent and conflictual elements is part of the exercise.

Establishing clear channels of communication and collaboration frameworks that align the objectives of various institutions is necessary to further mitigate conflicts between different levels of authorities, arising from competing priorities or lack of coordination, which can also impede co-creative processes.

On the other hand, it must be considered that “stakeholder involvement in scientific research may also undermine the perceived strength of the research, both in policy making and in scientific peer review” (Jolibert & Wesselink, 2012, p. 101). Technical difficulties, such as a lack of time, funding, and increased solicitations associated with emotional investment can constrain initiatives (Hinchcliff et al., 2014, p. 127).

In conclusion, while the challenges in stakeholder engagement are numerous, they are not insurmountable. By adopting inclusive, transparent, and adaptive approaches, promoting co-creation and trust, and overcoming technical and communication barriers, these challenges can be effectively mitigated, paving the way for sustainable and inclusive twin transition. In this respect, the following section on incentives and good practices highlight some possible directions.

5.2 Incentives and good practices documented in the literature

Incentives play a pivotal role in fostering meaningful stakeholder engagement (Jolibert & Wesselink, 2012). Researchers and practitioners emphasise that motivating stakeholders is essential for successful collaboration and achieving project goals. While the remuneration of stakeholders is viewed as a strong incentive (Forsythe et al., 2016; Mallery et al., 2012), other forms of incentives exist. As FITTER-EU allocated human resources only to the set-up of participation and stakeholder engagement, there is no budget available to compensate stakeholders through participation fees or covering travel and accommodation costs for attending in-presence events. Thus, unless partners individually decide to draw from their institutional budgets, FITTER-EU stakeholder engagement strategy will rely on non-material incentives, that are also highlighted as important in the literature and current practice and that can be clustered in few main types:

- Motivation to participate when own interests are directly affected. For example, stakeholders' participation may allow them to shape policies that affect their lives, or they may be contributing to the common good (Mallery et al., 2012). Providing incentives in this case is challenging, as the benefits are mostly intangible and/or lie in the future (Martinez et al., 2019a), and still this argument can and should be leveraged and used in communication as it could mobilise those who are motivated by (and can afford to) contributing to the common good beyond their own individual interest only.
- Opportunity to access to knowledge or training opportunities (Forsythe et al., 2016) is found to be useful. Generating products, such as handbooks and training manuals, that are valued and used by stakeholders can also be motivating (Hering et al., 2012).
- Access to data and project results, networking opportunities, gaining knowledge and skills, or increasing the visibility of their organisation (Groundtruth2.0 Project, 2018) can also be a motivating incentive.

Beyond incentives, recommendations and good practices can offer mitigation strategies for countering the likelihood of risks. These include the following:

- The stakeholder database should be a living document that grows throughout the project (Haeberlein & Hoevel, 2023); stakeholder engagement plans should also be revised in tandem with implementation (Martinez et al., 2019a).
- Contact stakeholders and begin sharing information with them as early as possible (Haeberlein & Hoevel, 2023; Jolibert & Wesselink, 2012).
- Clear aims, expectations and roles (Boaz et al., 2018; Forsythe et al., 2016; Haeberlein & Hoevel, 2023; Hering et al., 2012; Hinchcliff et al., 2014; Jolibert & Wesselink, 2012), as well as transparency (Concannon et al., 2014; TeRRIFICA, 2019).
- A recognition of stakeholder engagement as sustained, continuous, iterative and ongoing (Boaz et al., 2018; Forsythe et al., 2016; Hinchcliff et al., 2014; Jolibert & Wesselink, 2012).
- Identifying and allocating the necessary resources for stakeholder engagement (e.g., time, skills, funding) (Boaz et al., 2018; Haeberlein & Hoevel, 2023).
- Foresee and plan incentives for stakeholders (Jolibert & Wesselink, 2012) (Martinez et al., 2019a).
- Tailor the experience and methods of soliciting input to the specific stakeholder group (Mallery et al., 2012).

- Putting in place methods for evaluating stakeholder engagement (Boaz et al., 2018; Haeberlein & Hoevel, 2023).
- Embracing diversity in order to learn from and with stakeholders representing different interests, who can contribute a wide range of perspectives (Haeberlein & Hoevel, 2023; Szarka et al., 2023).
- Balancing strategic and pragmatic approaches (Kujala et al., 2022).
 - Strategic aspects include financial performance, risk management, and value creation; knowledge creation and learning; and reputation building. Strategy relates to embedding stakeholder engagement in a framework or model of research use and planning stakeholder engagement activity as part of the program of work (Boaz et al., 2018), selecting stakeholders strategically (Forsythe et al., 2016), and aligning expectations on costs (e.g., time and effort) and benefits (e.g., influence, access to results) (Jolibert & Wesselink, 2012).
 - Pragmatic aspects include context-dependent problem-solving and decision-making (Kujala et al., 2022). Flexibility also concerns being able to accommodate engagement and the outcomes of engagement in the research process (Boaz et al., 2018) and adapting to the practical needs of stakeholders (Forsythe et al., 2016).

5.3 Addressing challenges in the FITTER-EU project

Re-interpreting lessons learned from other projects and the literature on incentives for stakeholder engagement, to adapt them to the FITTER-EU project context, implies trying to answer a series of concrete questions:

- What are the specific challenges that could emerge per type of activity involving a stakeholder engagement component?
- What are specific mitigation measures?
- How can mitigation measures and incentives be customised to suit each case study country and stakeholder's type? (See Chapter 6)

On the part of researchers and R&I teams in FITTER-EU, several elements signal that they share a common view based on the importance and value of stakeholder engagement, and therefore have strong enough incentives:

- Dedicated tasks, tools and efforts across WP2 and WP7;
- A methodological approach based on systemic policy design and feminist intersectionality, and an overall methodology that integrates stakeholder engagement across all research and innovation tasks.

On the other hand, it is important to keep in mind that several factors could hinder the above-mentioned incentives for researchers. For example:

- De-prioritisation of stakeholder engagement as the research and analytical tasks become more demanding during the course of the project;
- Potential contradictions between research needs and engagement needs/expectations from participants. This is often the case when research questions that need to be answered have a scope or need to be formulated in ways that are not optimal for raising the interest of external

stakeholders. Research questions may also not always be suitable for hands-on activities that can represent valuable learning experiences for participants. This is also the case when research imposes data collection methods (i.e. audio/video recording) that are conflicting with the ease needed for group discussion activities.

The subsection below aims to map specific challenges per type of activity, and to outline corresponding mitigation strategies and incentives.

5.3.1 Specific challenges by activity cluster

Table 12: Likelihood of specific challenges per activity cluster

Challenges	Task clusters				
	1. Individual engagement activities for the development of the FITTER ecosystem	2. Group engagement activities for the development of the FITTER ecosystem	3. Sessions dedicated to validation of the FITTER-EU recommender system	4. Technological demonstrations with usability assessments	5. Engagement activities dedicated to dissemination, communication and exploitation
Dependence on key individuals: The project might be overly reliant on a few key stakeholders, posing a risk if they are unavailable	High	High	High	High	High
Availability in terms of time capacity, related to too many work responsibilities	High	High	High	High	High
Wrong communication channel chosen	High	High	High	High	High
Online format of participation challenges the group dynamic, limits the feeling of responsibility and reliable attendance	Medium	High	Medium	Medium	High
In presence workshops are more time consuming and require funding for travelling	High	High	None	None	High
Research is too critical towards current government strategies or neoliberal structures for involvement	Medium	High	Medium	Low	High
Conflicts of interest – stakeholders might have personal or professional interests that conflict with the project's objectives	High	High	Medium	Low	Medium

Challenges	Task clusters				
	1. Individual engagement activities for the development of the FITTER ecosystem	2. Group engagement activities for the development of the FITTER ecosystem	3. Sessions dedicated to validation of the FITTER-EU recommender system	4. Technological demonstrations with usability assessments	5. Engagement activities dedicated to dissemination, communication and exploitation
Communication barriers or miscommunication within the stakeholder group in terms of group imbalance and misunderstandings	Low	High	High	High	Low
Language barriers (since our project activities are conducted in English, as well as results, it excludes stakeholders who do not speak English fluently or who do not feel comfortable talking in English)	Medium <i>(this risk will depend on the final decision on methodologies in the diverse tasks)</i>	Medium	Medium	Medium	Medium
Availability in terms of time capacity related to stakeholder diversity (care work responsibilities, working part time, etc.)	High	High	High	High	High
Diversity of stakeholder group (in terms of people with diverging characteristics being underrepresented in the stakeholder type itself, sector specific)	High	High	High	High	High
Challenges in involving WINTA stakeholders (women, inter, non-binary, trans, agender) due to gendered barriers to participation	High	High	High	High	High

Challenges	Task clusters				
	1. Individual engagement activities for the development of the FITTER ecosystem	2. Group engagement activities for the development of the FITTER ecosystem	3. Sessions dedicated to validation of the FITTER-EU recommender system	4. Technological demonstrations with usability assessments	5. Engagement activities dedicated to dissemination, communication and exploitation
No digital skills to participate via online tool	Medium	Medium	Medium	Medium	Medium
Public sector actors are not open to bottom-up policymaking	High	High	Medium	Medium	Medium
Bureaucratic processes and institutional inertia interfere with engagement	Low	Low	Medium	Medium	Low
Lack of perceived profitability (for private sector and business angels)	Medium	Medium	Medium	Medium	Medium
Low visibility of the FITTER-EU profile discourages engagement	High	High	Medium	Medium	High
Difficulties in identifying/contacting sufficient number of relevant stakeholders	High	High	Medium	Medium	High
Contacted stakeholders fail to engage, or they drop engagement at some stage of the project, due to lack of interest or other constraints	Medium	Medium	Medium	Medium	Medium
Stakeholders disrupt engaging activities/events (e.g., co-creation workshops) due to deep-seated resistances	Low	High	Medium	Medium	High

Challenges	Task clusters				
	1. Individual engagement activities for the development of the FITTER ecosystem	2. Group engagement activities for the development of the FITTER ecosystem	3. Sessions dedicated to validation of the FITTER-EU recommender system	4. Technological demonstrations with usability assessments	5. Engagement activities dedicated to dissemination, communication and exploitation
Stakeholders attend planned events but fail to actively participate/contribute	Medium	High	Medium	Medium	Medium
The prioritised strategic stakeholders which have already participated in consultations on twin transition policies show no sufficient interest in the project	Medium	Medium	Low	Low	Medium
Within-country regional levels, imbalance in representation of stakeholders from different areas (e.g. urban/rural)	High	High	Low	Low	High
Difficulties receiving responses from organisations that partners haven't been in contact with before	High	High	High	High	High
Conflicts arise between different participants due to conflicting interests/views	Low	High	Medium	Medium	Medium
Lack of interest in the twin transition topic: Some stakeholders may see the green and digital transitions as irrelevant to their day-to-day operations, especially in sectors that are not directly affected.	High	High	Medium	Medium	High
Low expectations from engaging in the project: stakeholders may not believe that their contributions will have a meaningful impact on the project's outcome or policy decisions.	High	High	Medium	Medium	Medium

Challenges	Task clusters				
	1. Individual engagement activities for the development of the FITTER ecosystem	2. Group engagement activities for the development of the FITTER ecosystem	3. Sessions dedicated to validation of the FITTER-EU recommender system	4. Technological demonstrations with usability assessments	5. Engagement activities dedicated to dissemination, communication and exploitation
Resistance to change in traditional industries: Sectors like agriculture, fishing, and small-scale local businesses may be reluctant to adopt the changes required by green and digital transitions.	High	High	Low	Low	High

Dependence on key individuals: The project might be overly reliant on a few key stakeholders, posing a risk if they are unavailable. This challenge applies to all stakeholder types. Mitigation measures include contacting them early on, giving them a chance to brief and send representatives to FITTER-EU meetings. Utilising social media platform to share and engage other appropriate stakeholders may also be useful. Incentives include mentioning other participants, highlighting the possibility to network.

Availability in terms of time capacity, related to too many work responsibilities. This challenge applies to all stakeholder types, particularly CSOs, trade unions, research and academia, public sector and private sector (especially SMEs). Many organizations are likely to struggle with limited time, funding, or human resources to fully engage with the project. Mitigations include: ensuring that any tasks are accompanied by a deadline, shared well in advance, so that stakeholders can plan accordingly; plan shorter, more focused meetings where possible (e.g., 30 minutes around lunch breaks); offer possibilities for asynchronous participation (e.g., email feedback, surveys); plan events well in advance, sending out invitations in a timely manner; create a balanced mix of online and in-person events in order to match specific organisations/people with the type of event that works best for them; ensure the provision of refreshments in in-person meetings; align project events with academic calendars; and providing capacity building opportunities and resources, as well as access to digital tools, networking or webinars. Incentives include providing access to unique research data and collaboration opportunities. In addition, through our newsletter we can offer to collaborate in projects ran by stakeholders, should they have engagement needs.

Wrong communication channel chosen. Misaligned communication strategies, including using the wrong communication channels, can alienate stakeholders who may not receive relevant information. This may affect all stakeholder types. Mitigations include analysing the habits of stakeholders to identify their preferred channels. Monitoring the efficiency of different communication channels can also prove useful.

Online format of participation challenges the group dynamic, limits the feeling of responsibility and reliable attendance. This may affect all stakeholder types. Possible mitigation measures include making the format interactive, switching gears regularly between formal input and informal possibilities to exchange questions or ideas. Asking for the preferred video meeting platform and giving instructions on how to improve sound and video quality is also a possibility. Tailored social media outputs can be created after meetings or events, enabling participation to be publicly shared.

In presence workshops are more time consuming and require funding for travelling. Taking the time to meet in person fosters the connection between stakeholders and gives the possibility to engage more deeply in the possible multi-stakeholder dialogue on the twin transition. However, this challenge may affect all stakeholder types. Mitigation measures include choosing the locations of in-person strategically to facilitate stakeholder participation, minimising their effort. In this case too, tailored social media outputs can be created after meetings or events, enabling participation to be publicly shared.

Research is too critical towards current government strategies or neoliberal structures for involvement. This challenge is likely to mainly affect political actors and the private sector. Mitigation measures include finding out what their interest in the twin transition entails, and what are the points that they feel are too critical, in order to address these concerns. In addition, invitations can be formulated in slightly different tones towards different stakeholders. Environmental, social and economic issues, which are always interrelated, can be highlighted.

Conflicts of interest – stakeholders might have personal or professional interests that conflict with the project's objectives. This challenge is likely to affect CSOs, trade unions, researchers and political stakeholders. Mitigation measures include: setting an ethical frame for stakeholder dialogues, making meeting spaces safe for everyone and offering guidance and conflict resolution in case of conflicts of

interest; appointing a responsible person in meetings to deal with immediate issues, while highlighting our interest in stakeholder input, especially if they have valid and justified critique towards the project team and their perspectives; highlighting how a fair discussion can improve and accelerate the knowledge gain around twin transitions.

Communication barriers or miscommunication within the stakeholder group in terms of group imbalance and misunderstandings. This challenge can affect all stakeholder types. Feminist moderation and mediation strategies should be applied to ensure that all voices are heard and the potential imbalances in involvement are addressed. Providing a discrimination-free, safe space for all participants and clearly communicate that everyone is listened to. The opportunity to send private messages to the organisers via the chat function during calls or organise bilateral telephone calls to discuss possible problems calmly, are particularly useful tools.

Information overload, too much information, or information that is not well-targeted, can overwhelm stakeholders and reduce their ability to effectively contribute to the project. This challenge can affect all stakeholder types. A possible mitigation strategy is to reflect internally to the project on how to minimise or group information distribution, breaking down the complexity of the approach and highlighting the main project goals. Developing go-to explainer resources may also be useful.

Language barriers (since our project activities is conducted in English, as well as its results, there is a chance of excluding stakeholders who do not speak English fluently or who do not feel comfortable talking in English). This challenge can affect all stakeholder types. A possibility would be to offer automated translations or subtitles through digital platforms (e.g., within Zoom). In specific cases, we could also ask if an intermediary could participate in meetings to translate or facilitate communication, or to provide the expert with information afterwards.

Availability in terms of time capacity related to stakeholder diversity (care work responsibilities, working part time, etc.). This challenge can affect all stakeholder types, and especially those in caregiving roles. Meetings should be planned during core working hours, ensuring that participants are comfortable with selected times (e.g., by asking for preferred meeting times and places). Where possible, ensure that possibilities are present for children and/or older adults to be cared for.

Diversity of stakeholder group (in terms of people with diverging characteristics being underrepresented in the stakeholder type itself, sector specific). For example, challenges may arise with attempting to engage migrant and minority communities. This challenge can affect all stakeholder types. It will be important to give special attention to inviting targeted civil society organisations and providing incentives for their participation. Mitigation measures include: paying attention to the particularities of the given sector while contacting stakeholders, asking who is responsible for specific tasks and aiming to involve also practical experts (not merely the heads of organisations); if the stakeholder group is nevertheless very homogenous, acknowledge it during the meetings and point to the bigger picture; if linguistic barriers are there, make use of the project platform to provide translated materials and look into other translation possibilities; partner with local organisations that work with these communities to act as mediators and advocates.

Challenges in involving WINTA stakeholders (women, inter, non-binary, trans, agender) due to gendered barriers to participation. This challenge can affect all stakeholder types. Mitigation measures include paying attention to gender balance when choosing stakeholders to avoid all-men stakeholder groups or asking organisations directly to choose a WINTA person from their organisation, letting them know that we try to be flexible to attend diverse working schedules.

No digital skills to participate via online tool. This challenge can affect all stakeholder types, and particularly older stakeholders. Mitigation possibilities include: offering an introductory call to go through the different functions of the digital tool; offering alternative forms for providing input, such as via email or phone call; prioritise accessibility and clarity; where relevant, develop FAQs and help contents or video demos on how to proceed in different use cases.

Public sector actors are not open to bottom-up policymaking. In the case of public sector actors resisting the idea of participatory decision-making, advocacy and policy dialogue can be strengthened with the goal of building trust. A possible incentive can be provided by highlighting the potential for positive policy impact and offering recognition in publications.

Bureaucratic processes and institutional inertia interfere with engagement. This challenge may affect public sector stakeholders in particular. A mitigation option is to streamline processes and align project goals with public policy. A possible incentive could be to engage stakeholders through pilots or localised initiatives to demonstrate impact.

Lack of perceived profitability. This challenge could affect private sector stakeholders and business angels. A mitigation strategy could be to emphasise the long-term strategic benefits of the twin transition and of CSR. Incentives include networking opportunities and visibility in project outcomes.

Low visibility of the FITTER EU profile discourages engagement. This challenge may affect all stakeholder types. Fostering the visibility and reputation of the project and its partners on social networks and relevant channels focused on research and analysis of twin transitions is key. The FITTER-EU website should become a dynamic communication platform, reflecting the project's expected contribution in the domain of the twin transition, social equality and inclusion.

Difficulties in identifying/contacting sufficient number of relevant stakeholders. This challenge is likely to affect particularly public sector stakeholder, private sector stakeholders and business angels.

Contacted stakeholders fail to engage, or they drop engagement at some stage of the project, due to lack of interest or other constraints. This challenge relates to all stakeholder types but will be particularly relevant for private companies/business angels who rarely engage with issues of eco-social justice in relation to the twin transition. A close monitoring of stakeholder engagement throughout the project, through review meetings, could be beneficial. If a stakeholder drops engagement or fails to engage from the start, alternative contacts should be identified through existing networks.

Stakeholders disrupt engaging activities/events (e.g., co-creation workshops) due to deep-seated resistances. This challenge is likely to affect private sector stakeholders and employers' organisations. FITTER partners conducting this type of activities should be trained in how to deal with resistances in a multi-stakeholder setting.

Stakeholders attend planned events but failed to actively participate/contribute. This challenge can affect all stakeholder types. Post-event reports could include a dedicated section on participation and analyse the situation, with the aim of identifying the causes of this issue, and, if necessary, make improvements to avoid such situation is repeated in subsequent events.

The prioritised strategic stakeholders which have already participated in consultations on twin transition policies show no sufficient interest in the project. This challenge can affect all stakeholder types. A possible mitigation is to define and communicate a clear presentation of the project goals that emphasises inclusiveness and the aim of countering current risks. Another possibility is to better understand/explore

stakeholder needs and ensure the visibility of strategic stakeholders on project channels. A visibility/communication plan could be set up, aligning with project partners in charge of WP7.

Within-country regional imbalance in representation of stakeholders from different areas (e.g. urban/rural). This challenge can affect all stakeholder types. A possible mitigation strategy is to ensure that we reach out to a geographically balanced sample of organisations.

Difficulties receiving responses from organisations that partners haven't been in contact with before. This challenge can affect all stakeholder types. It is necessary to provide a clear email template with objectives and requirements, which will serve as a motivating and interesting presentation of the project.

Conflicts arise between different participants due to conflicting interests/views. This challenge can affect all stakeholder types. A possible mitigation strategy is to provide a structured agenda for each event; ensuring that facilitating skills are up to date.

Lack of interest in the twin transition topic. Some stakeholders may see the green and digital transitions as irrelevant to their day-to-day operations, especially in sectors that are not directly affected. This challenge could affect the public and private sector. Mitigation options include: designing awareness campaigns that link the twin transitions to immediate and tangible benefits, such as cost savings or new business opportunities; and using targeted case studies and real-life examples from similar sectors to demonstrate the value of participation.

Low expectations from engaging in the project. Stakeholders may not believe that their contributions will have a meaningful impact on the project's outcome or policy decisions. This also ties in with public distrust in government or institutional processes: low levels of trust in government decision-making may discourage engagement with the project. This challenge could affect CSOs, private sector (especially SMEs), and public sector stakeholders. Mitigation measures include: providing clear, transparent updates on how stakeholder input will be used and demonstrate examples of previous projects where engagement led to concrete results; and establishing a communication strategy so that participants can see how their input influences decisions, fostering a sense of ownership and contribution.

Resistance to change in traditional industries. Sectors like agriculture, fishing, and small-scale local businesses may be reluctant to adopt the changes required by green and digital transitions (especially in relation to private sector stakeholders). Mitigation options include: providing training materials and other information to promote the adoption of eco-friendly technologies and practices; and engaging experts to demonstrate how the twin transition can create economic opportunities and long-term sustainability.

6 Engagement strategies per stakeholder type

This chapter outlines engagement strategies for the various types of stakeholders prioritized by the FITTER project, including CSOs, public service/ policymaking, employer's organisations, business angels, stakeholders from the private sector, trade unions and Research/Academia. The focus is on connecting the engagement procedures described in Chapter 4 with the national action plans (NAPs).

The chapter is divided into two sections. The first Section addresses general strategies for engaging the prioritized stakeholders categorized in the NAPs elaborated by partner for the six country case studies. The second Section provides a detailed examination of the national action plans, offering specific strategies for each type of stakeholder and country.

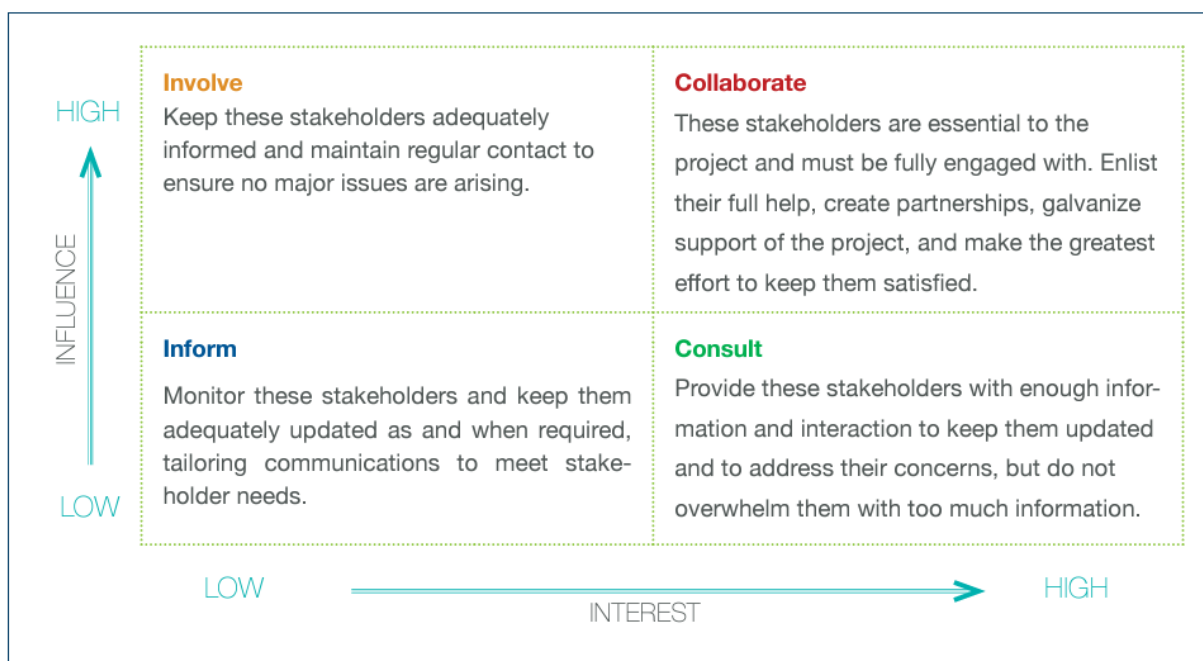
6.1 General strategies with the prioritised stakeholders

Stakeholder appraisals have been conducted by FITTER-EU partners in the NAPs based on their potential degree of engagement, using two main dimensions derived from the work of Mendelow (1991). Mendelow's framework prioritises stakeholders and defines engagement strategies by evaluating two key factors: the level of interest stakeholders have in the project and the influence they may exert on the project's outcomes.

Mendelow proposes an influence-interest grid to classify stakeholders targeted by a project, enabling the development of tailored engagement strategies (Durham et al., 2014; Hollmann et al., 2022).

This author proposes an influence-interest to classify stakeholders targeted by a project to develop specific engagement strategies (Figure 2).

Figure 2: Mendelow's influence-interest grid



Source: Durham et al., (2014)

In particular, different types of actions are recommended for each category:

- Key stakeholders (high-influence and high-interest): the objective with this group is to establish collaboration. It is generally recommended to start engagement at beginning of the project and establish continuous communication mechanisms.
- Influencers: (high-influence and low-interest): they might be key actors for the project; therefore, it is relevant to keep their needs satisfied. In addition, communication efforts might be worth to inform them about the project's benefits and raise their interest.
- Interested stakeholders (low-influence and high-interest): it is recommended to keep this group informed and establish some consultations about some specific areas of the stakeholders' interest.
- Passive stakeholders (low-influence and low-interest): Mendelow recommends to minimise engagement efforts with this stakeholder group.

6.1.1 Reinterpreting the Mendelow matrix in the FITTER-EU project

The FITTER project aims to develop a recommender system to support policymakers in ensuring fairness during the green and digital transitions. Thus, the “influence” dimension is closely related to a stakeholder's impact on policy processes and structures within a country concerning these transitions, or individually if the concept does not apply.

However, it is important to consider that there may be stakeholders who, while not directly involved in formal policy processes and structures, still propose innovative solutions relevant to the transitions. For example, grassroots climate movements may offer valuable input for the FITTER project despite not having direct influence within formal policy structures. If the “influence” dimension is strictly interpreted as direct participation in formal policy processes and structure, such stakeholders might be classified as having “low influence” and thus deemed irrelevant for involvement in the project.

To avoid excluding significant stakeholders, the “influence” dimension is being reinterpreted and expanded in FITTER-EU so as to ensure inclusion of actors who might follow a more critical or anti-categorical approach or who do not have enough power/resources/visibility to influence policy debates/processes. In particular, influence on the formal policy processes is being considered along with another dimension namely:

- Stakeholder's influence in public policy/political/media debates, as for the case of stakeholders which might be implementing alternative forms of mobilisation and advocacy.

More details on the procedures proposed for inclusiveness in the project can be found in Chapter 4.

As described in Chapter 4, the FITTER project's engagement strategy adopts a dual approach with both bottom-up and top-down components. The top-down component began with the initial mapping of stakeholders in April 2024 and is currently ongoing. As a result, FITTER-EU partners have conducted an initial classification of the mapped stakeholders using the Mendelow matrixes included in the NAPs. It is important to note that this initial positioning may be adjusted as stakeholders are contacted and as their relevance and interest in the project evolve.

The bottom-up engagement approach will commence in October 2024, involving direct contact with stakeholders and inviting them to join the FITTER-EU project. The “collaboration” section of the project's webpage will play a central role in this approach, as stakeholders can express their interest to become an

active part of the project by filling an online. All stakeholders who submit the webpage's form will be considered as having high interest in the project.

In what follows, some general considerations for the engagement of stakeholders are presented according to their position in the Mendelow matrix.

6.1.2 General recommendations for the engagement according to stakeholder's classification in the Mendelow matrix

Stakeholder engagement efforts should be concentrated on those classified as “high influence” due to their critical role in providing quality inputs for the development of the FITTER ecosystem and their potential importance for the future implementation of the platform. Within this group, engagement strategies should be tailored based on their interest in participating in the project.

Key stakeholders (high-influence and high-interest)

Stakeholders with high influence who are also highly interested in participating in the FITTER-EU project might become potential “partners” in the engagement process. They are significant as they could have a relatively stable role in being involved in the participatory activities of the FITTER methodology and be pivotal in the project's diffusion and establishing new contacts. Also, they are less likely to experience the “engagement fatigue” risk. Therefore, they could have a (relatively) stable role in most project activities belonging to Clusters 1 and 2 (see Chapter 4). Some recommendations for engaging with these stakeholders:

- Target these stakeholders as the first actors to be informed about the “bottom-up” engagement path. Confirm their status as part of a “key stakeholders” group if they express willingness to become actively involved in the project across its different stages.
- Conduct an internal evaluation —integrated with all national partners in the case of Ireland, Spain and Italy— to determine whether some of these stakeholders could be considered as “key partners” with a central role in the stakeholder engagement strategy. If so, it may be beneficial to arrange a dedicated meeting with the selected stakeholders to gauge their willingness and potential for partnership.
- Consider involving prioritised stakeholders to provide feedback/input on the FITTER-EU methodology and its unfolding (with a stable participation) since the beginning of the process, instead of inviting them into isolated activities only.
- These actors might play an important role in establishing new contacts. Explore this possibility with them from the initial contact and, while paying attention that this doesn't limit the inclusiveness of the process.
If they are inclined towards a more active role, assist them in developing and adapting the project's communication and diffusion materials, as they might have an interest into communicating that they are participating to the FITTER-EU Research process.

Influencers (high-influence and low-interest)

Like “key stakeholders,” “influencers” are crucial to FITTER-EU in achieving the project's objectives. However, engaging these stakeholders can be more challenging due to their low interest in the project. Consider the following strategies for their involvement:

- Involving “influencers” requires significant effort. Begin engagement efforts at the start of the project, ideally when the bottom-up engagement strategy commences in October 2024. Avoid last-

minute communications or invitations with short notice, as these may not be effective with this group.

- It is essential to use clear and effective communication with this stakeholder group. Clearly outline the FITTER-EU objectives and methodology in your messages. Invitations should specify: (i) their specific contribution to the project and their expected role in the activity, and (ii) how they would benefit from participating.
- “Influencers” may be more susceptible to engagement fatigue, so excessive contact could be counterproductive. To prevent “stakeholder burnout,” identify two or three key activities where their involvement is particularly valuable before making contact. Communicate this intention in the invitation and adjust your plan based on their feedback.
- Specific contributions of these stakeholders might be particularly valuable for activities belonging to Clusters 1,2 and 3 (see Chapter 4) —individual and collective activities for the development of the FITTER ecosystem, as well as sessions to validate the recommender system, respectively—. Avoid involving them in more general or massive activities, such as the ones belonging to Clusters 4 and 5 —technological demonstrations (hackathons) and diffusion and communication and exploitation activities—.
- Refer to Chapter 5 for more details on the challenges related to the involvement of stakeholders and to explore potential mitigation measures and incentives to make engagement smoother.

Stakeholders with a low degree of influence (interested and passive stakeholders)

“Interested” and “passive” stakeholders are positioned in the two quadrants of the Mendelow matrix that reflect a low degree of influence. While “interested” stakeholders are highly engaged with the project, “passive” stakeholders are not. Consider the following for their involvement:

- Given the substantial number of stakeholders needed for the project's participatory activities, the importance of “interested” stakeholders should not be underestimated. Their involvement can provide significant diversity and valuable perspectives.
- Include “interested” stakeholders in larger activities, such as those in Clusters 4 and 5, which involve a broader range of participants
- Although “interested” stakeholders might be more flexible regarding short-notice invitations compared to “influencers”, it is still advisable to adhere to the timing guidelines for engagement outlined in Chapter 4.
- Even though “passive” and “interested” stakeholders have a low level of influence, it is beneficial to include them in the bottom-up engagement path of the FITTER-EU project. However, priority should be given to “key” stakeholders and “influencers” in this engagement process.

As a final consideration, these recommendations are intended as guidelines and should be adapted to the specific national and local contexts. Additionally, a stakeholder's position in the Mendelow matrix may shift throughout the project's duration. Therefore, partners could periodically use the matrix as a tool to evaluate the status of stakeholder engagement and adjust strategies based on changes in influence and interest dimensions.

It is worth considering that the initial stakeholder positioning in the NAPs represent an initial analytical proposal based on partners’ assessments. Despite this, once stakeholders will be contacted with the start of the bottom-up engagement path in October 2024, their actual level of interest will be revealed, which may necessitate adjustments to the initial engagement strategies.

The following Section provides more specific strategies for each type of stakeholder and country, based on the information included in the NAPs.

6.2 Strategies per type of stakeholder

6.2.1 CSOs

CSOs are one of the stakeholder groups targeted by the FITTER project, which are more likely to be interested in participating.

Key inputs of the comparative analysis

- **CSO are favourable for the engagement:** CSOs are one of the most active and favourable stakeholders for the engagement, because of their high level of activity regarding twin transition issues and their focus on structurally vulnerable groups who are affected by transition policies.
- **Common themes addressed by mapped CSOs:** are environmental and social justice, advocacy for structurally vulnerable populations, and pushing for more inclusive and sustainable policies.
- **Limited formal influence:** in most of the project countries, the influence of CSOs in policy processes is limited, with only a subgroup of these organisations actively participating in formal policy processes and structures. In Germany, CSOs directly participating in policy structure and processes are mainly large umbrella organisations such as alliances and federations which are considerably influential depending on the sector. In Hungary their participation in policy processes is very limited due to the centralisation of the government, while in Italy CSOs are increasingly marginalised in policy processes related to the RRF. The exception to this situation is Ireland, where policy processes are highly participatory and CSOs actively participate: they are well-integrated into participatory policymaking frameworks, such as Citizens' Assemblies, providing a formal mechanism for influence.
- **CSOs usually adopt critical and anti-categorical approaches, both in relation to climate and social issues, and to the interests of the structurally vulnerable population groups they might be representing:** this is the case particularly in Hungary and Italy, where stakeholders have a relevant role in the monitoring and critically following-up public policy processes. Likewise, Spanish CSOs are particularly active in the regional and local levels, usually having a critical approach towards national policy.
- **Most of the influence of CSOs in the policy debate and public opinion is through advocacy, grassroots mobilisations and independent initiatives outside policy processes.**

Recommendations for the engagement

1. **Tailor messaging to align with CSO objectives.** Clearly communicate the project's goals and the relation with strategic objectives of organisations, such as social justice, structurally vulnerable groups, environmental protection, etc.
2. **Provide tangible benefits, such as capacity-building, opportunities to increase visibility and networking.** Highlighting these advantages will help CSOs with limited resources engage more fully.
3. **Maintain transparency to build trust, especially in more centralised political contexts.** Maintain clear, transparent communication about the scope of participation, expected outcomes, and how CSOs' contributions will influence the project. Lack of trust in formal processes could be a barrier for participation.

4. **Highlight networking and knowledge-sharing opportunities.** Emphasise the potential for CSOs to expand their networks and access knowledge from international or cross-sectoral collaborations.
5. **Offer flexibility in participation formats.** Customise the involvement into FITTER-EU participatory tasks according to the different capacities and levels of involvement of different organisations. Flexible participation formats help reduce the burden on CSOs with fewer resources.
6. **Provide early and comprehensive background information.** Share relevant information about the project's goals, scope, and potential outcomes well in advance to allow CSOs to prepare and assess alignment with their own objectives.
7. **Leverage advocacy on twin transitions and social justice.** Emphasise the project's role in advancing both green and digital transitions in ways that promote social justice and equity, particularly focusing on how structurally vulnerable groups are addressed. CSOs across all countries, especially in Ireland and Spain, are advocating for more inclusive transition policies that address inequality. Positioning the project as aligned with these goals will resonate strongly.

6.2.2 Public service, policymaking

Key inputs of the comparative analysis

- **Levels of centralisation in governance structures managing the twin transitions:** policymaking structures related to the twin transitions differ among the FITTER-EU countries, with highly centralised countries, such as Hungary and Italy, and others, such as Germany and Spain, which offer more space to regional and local governments in adapting national policies as part of their governance structures. Ireland, though, is the country with the most inclusive governance structures, with formally participatory mechanisms established.
- **Focus on economic governance vs social and environmental justice:** these are common topics of political debate in the project countries, such as Germany. Some countries, such as Hungary, prioritise economic objectives.
- **Challenges in engaging policymakers:** time constraints, competing priorities, and bureaucratic hurdles make it difficult to involve senior policy officers, particularly at the national level. It might be easier to work with the local and regional levels of government.

Recommendations for the engagement

1. **Align the communication of the project's objectives with national priorities and policy agendas.** Policymakers are more likely to engage if they see alignment with national or regional priorities. Try to adjust the communications with the policy debates, considering the tensions there might be between the economic and social and environmental development.
2. **Provide evidence-based insights that support decision-making processes and strategic planning.** Offering concrete, data-driven insights that can support policymakers in shaping legislation or strategic planning might be a relevant incentive for policymakers to join the FITTER-EU project. When engaging policymakers, refer to the anticipated access to project results as one of the incentives for this stakeholder group.
3. **Engage local and regional authorities.** Focus the involvement of senior civil servants and policymakers from the national level of government in concrete contributions which requires highly specialised inputs (mainly activities from Clusters 1,2 and 3). Local and regional stakeholders might be more open to participate in more than one activity of the project or to have a more stable participation if they get involve with the project.
4. **Emphasise EU-level relevance to tap into European frameworks and funding opportunities.** Emphasise the project's connection to broader EU policies and frameworks, particularly those

related to the Green Deal, Just Transition, or Digital Europe. Showing the project's European context can make it more attractive to national public service stakeholders.

5. **Offer flexible engagement opportunities tailored to the time constraints of public service stakeholders.** Tailor engagement to their specific roles and responsibilities to ensure it is relevant to their work.
6. **Foster transparency to build long-term trust and credibility with government actors.** Be transparent about the scope of participation, the expected outcomes of the project, and how public service stakeholders' contributions will be used. Building trust and clarity since the beginning of the engagement might increase the probability for long-term cooperation.
7. **Provide public recognition and visibility for participation to increase motivation.** Offering visibility through project outputs, reports, and media channels might be of specific interest of politicians belonging to small parties.
8. **Address specific policy challenges by tailoring the project to offer relevant, actionable solutions.** Present the project as a tool that can provide targeted solutions the specific challenges that policymakers are facing.

6.2.3 Private sector, employers' organisations and business angels

Engagement strategies of these three stakeholder categories are being grouped due to the similarities in their incentives and motivations, led mainly by their private-sector interests, giving priority to economic profitability and their sectoral specificity.

Key inputs of the comparative analysis

- **Actors in these three stakeholder categories possess business-driven motivations for engagement.** However, divergences arise in the level of formal consultation, sectoral representation, and political alignment, influencing their potential involvement in the FITTER project.
- **Across all countries, time and resource constraints often limit the willingness of private sector and employer's organisations to engage in non-business-oriented projects like FITTER.**
- **In countries like Germany and Spain, SMEs are the widest type of organisation in the private sector.** However, usually the political influence of bigger companies and corporations is wider.
- **Large corporations and employer's organisations usually influence policy processes by lobbying activities and, in some cases, through the participation in formal governance participatory mechanisms.** On the other hand, business angels do not usually participate in formal policy processes or structures; however, they broadly focus on innovation, social impact and sustainability projects, aligning closely to twin-transition goals and the FITTER-EU project's objectives.
- **Actors belonging to these three categories tend not to adopt critical or anti-categorical perspectives.** In Spain, though, private sector stakeholders, particularly in agriculture and rural SMEs, have expressed critical views about the government's energy transition policies.
- **While business angels in all countries advocate for improved start-up policies, the level of policy engagement varies.** Business angels in Hungary have a particularly small community.
- **Stakeholders belonging to these three categories are more likely to engage with the FITTER-EU project if the project presents clear opportunities related to their organisational scope.**

Recommendations for the engagement

1. **Align the engagement message with stakeholders' interests, using their “professional slang”.** For instance, in the case of employer’s organisations align with advocacy agendas, demonstrating how the project supports employer organisations’ goals of workforce development and business competitiveness.
2. **Leverage corporate social responsibility (CSR) and reputation building, showing how participation enhances social and environmental credibility.** Prioritise the engagement with private organisations which might be addressing the effects of the twin transitions, such as corporations in the energy and technology sectors which might be active in climate change action.
3. **Appeal to organisations which might be interested in social impact.** Leverage the growing interest of private stakeholders by showcasing how the project contributes to inclusivity, social and environmental justice, or tackling societal challenges.
4. **Offer networking and collaboration opportunities with other sectors and stakeholders to foster innovation and partnerships.** Present the project as a platform for networking with a broad range of stakeholders, including innovative startups, research institutions, and policymakers, among others.
5. **Showcase technological advancements, positioning the project as a hub for cutting-edge innovation.** Emphasise the innovative potential of the recommender system which is being developed in the project, the FITTER-EU platform.
6. **Offer visibility for the participation of the organisations in the project, showcasing their involvement in sustainable and innovative initiatives and networks.**

6.2.4 Research/academia***Key inputs of the comparative analysis***

- **Research and Academia shape policies providing inputs to policy structures, providing evidence-based reports and through an advisory role.** In some countries like Germany, they have a formal role in policymaking, while in other like Hungary, they are marginalised.
- **Research institutions in some countries focus more on technical aspects of the twin transitions (e.g., Germany), while others emphasize the social impacts (e.g., Ireland and Spain).**
- Study of the twin transitions has a central role in research institutions, and they could be motivated by opportunities for collaboration, knowledge dissemination, and increasing their academic visibility.
- **They usually keep a critical perspective towards economic growth models underpinning current twin-transition policies.**

Recommendations for the engagement

1. **Link the project to policy debates and societal impact, showcasing how academic research can directly influence policymaking on twin transitions.**
2. **Provide platforms for knowledge dissemination and networking, giving academics the chance to present their research and engage with peers.**
3. **Highlight interdisciplinary collaboration, appealing to researchers from both technical and social science fields.**
4. **Connect research to practical policy applications, showing how academic work can directly shape real-world policies and practices.**

5. **Leverage EU-level research and innovation agendas, aligning the project with key European initiatives to attract funding and collaboration opportunities.**

6.2.5 Trade unions

Key inputs of the comparative analysis

- **In all of the FITTER-EU countries, trade unions are focused on worker's rights, with specific issues such as job security, fair wages, and working conditions in industries.** Their focus on the green and digital transitions relies on their effects on the workforce.
- **Trade unions usually implement critical approaches towards capitalism, usually keeping a traditional anthropocentric view focused on the rights of workers related to the new policy changes.** Less focus is commonly given to exclusively green aspects.
- **Trade unions are engaged in social dialogue with government and employers; however, their level of influence changes across countries.** In all of the countries with the exception of Hungary, where they are considerably marginalised, they formally participate in the policy processes.

Recommendations for the engagement

1. **Align with trade unions objectives for their engagement, workers' rights and social justice, emphasising how the project supports fair transitions and protects structurally vulnerable workers.** Mention that the project supports representing workers' participation in decision-making, ensuring their voice in shaping labour policies during transitions.
2. **Implement platforms for social dialogue, allowing trade unions to influence policy and engage with national and EU-level decision-makers.**
3. **Highlight fairness in the distribution of the benefits and burdens of transitions.** Emphasise how the project will ensure fairness in the distribution of both the benefits (e.g., job creation, innovation) and the burdens (e.g., job displacement, retraining needs) of the twin transitions. Position the project as a tool for ensuring that structurally vulnerable workers are not disproportionately affected.
4. **Provide concrete evidence of project benefits for workers, using data and case studies to demonstrate how the project could potentially improve working conditions and job security.**
5. **Engage unions early in the project design, involving them in planning to ensure their concerns are integrated and foster a sense of ownership.**

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7 Annexes

Annex N° 1: Stakeholder's database

The integrated version of the stakeholder's database is storage in the FITTER project's Sharepoint and available at the following [link](#). The path to find the document in the repository is: "Documents/General/WP2 – Conceptual framework and methodology / T2.3 FITTER expert group engagement strategy /Stakeholder database".

In the same folder of the Sharepoint the given [guidelines to fill the database](#) are included, which were sent to partners on July 17th as part of the elaboration process of the NAPs.

The following table details the dictionary of the variables included in the database.

Table 13: Dictionary of variables

Column	Variable	Description	Values	Notes
A	Country	In what country is located the organisation?	Select from a drop-down list	
B	Organisation name	What is the name of the organisation?	Open question	
C	Organisation acronym	What is the acronym of the organisation?	Open question	
D	Website	Include the organisation's website	Open question	
E	Organisation type	Considering the types of stakeholders targeted by FITTER, to what type of organisation	Select from a drop-down list	
F	Description/area of work		Open question	
G	Geographical scope	Does the organisation work/focus on predominantly at the local, regional, national or European level?	Select from a drop-down list	
H	Disadvantaged groups	Do not fill when adding new stakeholders		
I	Vulnerable groups	Do not fill when adding new stakeholders		
J	Focus on a vulnerable population group	Does the stakeholder organisation focuses on/represent the interests of vulnerable or disadvantaged population groups? (yes/no)	Select from a drop-down list: yes/no	New variable
K	Prioritised vulnerability (fill only if column J is "yes")	If the organisation focuses on vulnerable population groups (only if the answer to the previous question is "yes"), please detail in which	Open question	New variable
L	Contribution to intersectional approach	Would you consider that engaging this stakeholder in the FITTER project would contribute to have an intersectional approach? If so, how and with reference to which groups?	Open question	
M	Focus on energy sector	Does this stakeholder focuses in the energy sector? (yes/no)	Select from a drop-down list: yes/no	New variable

N	Focus on transport sector	Does this stakeholder focus in the transport sector? (yes/no)	Select from a drop-down list: yes/no	New variable
O	Focus on building & housing	Does this stakeholder focus in the building & housing sector? (yes/no)	Select from a drop-down list: yes/no	New variable
P	Focus on food and agriculture	Does this stakeholder focus in the agriculture sector? (yes/no)	Select from a drop-down list: yes/no	New variable
Q	Focus on other sectors	Does the stakeholder focus in other sectors?	Open question	New variable
R	Main sector	Do not fill when adding new stakeholders		
S	Additional sector(s)	Do not fill when adding new stakeholders		
T	Work on transitions	Does the organisation work/focus on the green, digital or twin transitions? Consider that organisation might use a different framing of the transitions	Select from a drop-down list	
U	Level of focused work on transitions	How would rate the focus of the organisation in the transition?	Select from a drop-down list	
V	Role in policymaking structures, processes and public debate	Is the mapped stakeholder an institutional or administrative actor or does it participate in governance structures or processes related to the twin transitions, or is it present in public policy debates through communication/mobilisation? If yes, provide some details	Open question	
W	Role in policymaking processes	Do not fill when adding new stakeholders		
X	Partner including the stakeholder in the database	Select the name of your organisation	Select from a drop-down list	
Y	Name of contact		Open question	

Z	Surname of contact	Fill with contact information of the organisation	Open question	
AA	Position of contact		Open question	
AB	Email address		Open question	
AC	Telephone number		Open question	
AD	Additional notes		Open question	
AE	Potential tester	Does the stakeholder have the potential to be one of the testers of the FITTER platform when this will be developed?		New variable
AF	Task(s) where contribution would be valuable	In what specific project WPs or tasks would it be worth involving this stakeholder?		New variable
AG	Initial contact status	Was the contact with this stakeholder already active at the moment when it was included in the database? Answer "active" if yes, "cold lead" whether it was an old contact or "not established"	Select from a drop-down list: active, cold lead, not established	New variable
AH	Initial frequency of contact	How frequent was the periodicity of contact with this stakeholder at the moment when it was included in the database?	Select from drop-down list: frequent/occasional/one-time/none	New variable
AI	Date added to the database	Include the date when the stakeholder was included in the database	Use the format dd/mm/yy	New variable
AJ	Already contacted for the project	Have the stakeholder already been contacted by the project?	Select from a drop-down list: yes/no	New variable. To be filled only when the actual engagement starts (after September). Only fill is the stakeholder has been contacted for T3.1
AK	Declared interest in participating	Has the contacted stakeholder declared interest in participating in the project? [only for stakeholders that has already been contacted]	Select from a drop-down list: yes/no	New variable. To be filled only when the actual engagement starts (after September). Only fill is the stakeholder has been contacted for T3.1

AL	Stakeholder filled the webpage form	Has the stakeholder filled the involvement form included in the project's webpage? [only for stakeholders that has already been contacted]	Select from a drop-down list: yes/no	New variable. To be filled only when the actual engagement starts (after september) and the collaboration section of the webpage is active
AN	Task participation	Please, mention the tasks in which this stakeholder has participated [only for stakeholders that has already been contacted]	Open question	New variable. If the stakeholder participated in T3.1 include "T3.1" in the column
AO	Any other comments on their engagement and participation	Include any relevant comments that the contacted stakeholder might have given related to the participation in the project FITTER, such as their availability of interest related to specific activities/initiatives [only for stakeholders that has already been contacted]	Open question	New variable. To be filled only when the actual engagement starts (after september). Only fill if the stakeholder has been contacted for T3.1

Annex N° 2: National action plans

The template with the guidelines provided to partners on July 17th 2024 for the NAPs can be found in the following [link](#).

They are included in the project's repository in the following path: "Documents/General/WP2 – Conceptual framework and methodology / T2.3 FITTER expert group engagement strategy /National action plans".

National action plans

- [Germany](#)
- [Hungary](#)
- [Ireland](#)
- [Italy](#)
- [Portugal](#)
- [Spain](#)

Annex N° 3: Initial developments to operationalise the bottom-up stakeholder engagement path

The present annex includes the information and communication material elaborated to activate the "collaboration" section of the FITTER-EU webpage, implemented in collaboration with UNINOVA, partner developing the webpage. In particular, the following materials were originally prepared and further adapted by UNINOVA in the webpage.

- **Text to be included in the "collaboration section of the webpage"**

Join the FITTER-EU Community of Stakeholders

Do you have expertise in the sectors of transport, energy, building and housing, or food and agriculture? Do you represent the interests of vulnerable groups? Are you interested in contributing to a fair green and digital transition? The FITTER-EU project is assembling a vibrant community of stakeholders, and we want to hear your perspective.

We are looking for stakeholders in the project case-study countries (Ireland, Italy, Spain, Germany, Hungary, and Portugal), as well as transnational experts with an EU-level perspective. The community will include expert researchers, policymakers and practitioners, as well as civil society organisations, trade unions and employer's organisations.

Stakeholders will be contacted to engage in different FITTER-EU activities and initiatives, including workshops, interviews, and roundtable discussions. Your contribution will inform research into enabling anticipatory governance to support a fair and inclusive twin transition in Europe.

If you want to network with like-minded experts across Europe and contribute to forecasting strategies for a sustainable future, sign up using the form below!

- **Infographics**

Figure 3: Infographic for CCGS involvement – First version (elaborated by SV)



Figure 4: Infographic for CCGS involvement – Second version (adapted by UNINOVA)



- **Registration form for the bottom-up engagement (operationalised in the webpage)**

The registration form for the bottom-up engagement has been operationalised by UNINOVA in the FITTER-EU webpage according to the following fields indicated by SV. The variables (information fields) are aligned to the variables included in the Stakeholder's database, with the intention to allow a smooth integration of the new contacts registered in the form in the stakeholder's database.

First name* (Open question)

Last name* (Open question)

E-mail* (Open question)

Telephone (Open question)

Organisational affiliation (name of the organisation)* (Open question)

Country of the organisation* (choose one of: Italy; Ireland; Spain; Germany; Portugal; Hungary; Other - please specify)

Role within the organisation (Open question)

Organisation type* (Choose one of: Civil society organisation; Employers' organisation; Private sector; Public service/policy-making; Research/academia; Trade union; Business Angel; Other-please specify)

Brief description of your organisation (Open question)

Geographical scope of your organisation* (Choose one of: Local, Regional, National, European, Other – please specify).

Please select the sector(s) in which your organisations focus* (Choose one or more of: Building and housing; Food and agriculture; Energy; Transport; Other – please specify)

Do you and/or your organisation work on the digital, green or twin transition?* (Choose one: Digital transition; Green transition; Twin transition; None of the above).

Do you and/or your organisation work with and/or represent vulnerable groups?* (Choose one option: yes/no. If yes, open field to specify)

Does your organisation follow an intersectional approach? If so, how? (Open question)

Does your organisation participate in governance structures or processes related to the twin transitions, or is it present in public policy debates through communication/mobilisation? If yes, provide some details (Open question).

I have read and agree with the Stakeholder Database consent form of the FITTER-EU project. I consent to my data being processed for the purposes of the FITTER-EU project, and to being contacted about and invited to project activities and events.* (Checkbox)

* Required fields

- **Invitation to join the FITTER-EU project**

Dear *[insert name and position inside the organisation]*

It is a pleasure greeting you on behalf of the FITTER-EU project [\[insert link of the project's webpage\]](#). We invite you to become a member of the project's Cluster Community Group of Stakeholders (CCGS) and contribute to an innovative project aimed at fostering a fair and inclusive twin —green and digital— transition in Europe.

The FITTER-EU project, funded within the Horizon Europe Programme, is dedicated to researching the origins, dynamics, and determinants of inequalities related to the twin transition, with particular focus on the six countries of the project: Germany, Hungary, Ireland, Italy, Portugal and Spain. The project studies the effects of the transitions across four sectors —Energy, Transport, Building & Housing, Food & Agriculture — and the policy responses that support decision-making for a just twin transition in Europe. The final outcome of the project is the development of the FITTER Ecosystem, a highly interactive and gamified digital platform powered by an Intelligent Predictive Decision Support System. This platform will inform and provide inputs to policymakers regarding the effects of the twin transitions in prioritised population groups at-risk of suffering their adverse effects.

Joining the CCGS

By joining the CCGS, you will have the opportunity to participate in the FITTER-EU methodology, helping to shape the FITTER Ecosystem, contributing to the analysis of the effects of the twin transition on the inequality patterns, as well as proposing effective solutions to promote social justice and inclusivity. By joining the CCGS you will be able to:

- Network with key stakeholders, including policymakers, civil society organisations, trade unions, employer's organisations, business angels and actors from the private sector and public institutions.
- Gain visibility and contribute to cutting-edge research on inequality and social justice during the green and digital transitions.
- Play a critical role in shaping policy recommendations that ensure an equitable twin transition
- Provide input that will directly influence the development of the FITTER Ecosystem, contributing to the decision-making to tackle the adverse social impacts of the twin transition.
- Test the FITTER Ecosystem and contribute to their validation.

We invite you to express your declaration of interest in joining the CCGS by filling the form available at our webpage [\[insert link\]](#). Upon your non-binding expression of interest, you will receive updates on the project's development and our team will reach out to invite you to participate in the activities of the project's methodology more suitable to your professional profile.

Your expertise and engagement will allow the successful implementation of the FITTER project. We look forward to welcoming you to the FITTER community!

Best regards,

[Signature]

- **Informed consent form on registering to the FITTER-EU CCGS /database (to be updated)**



Research information for informed consent (STAKEHOLDER DATABASE)

FITTER-EU: Fair and inclusive twin transitions for a stronger social Europe

1. Introduction

FITTER-EU is a project funded by the European Commission under the Horizon Europe Framework Programme (grant agreement No 101132546). By signing up to the FITTER-EU Stakeholder Database, you may be contacted by researchers as part of the project, inviting you to participate in the project's stakeholder engagement activities.

Before you decide whether or not to sign up, it is important for you to understand why the research is being done and what it will involve. Please take time to read the following information carefully.

2. Who is responsible for the research project?

All the project partners are from EU Member states. The FITTER-EU project is co-ordinated by TECHNOLOGICAL UNIVERSITY DUBLIN (TU Dublin) (Ireland). The consortium is additionally composed of:

- TFC RESEARCH AND INNOVATION LIMITED (TFC) (Ireland);
- SMART VENICE SRL (SV) (Italy);
- AITEC ASESORES INTERNACIONALES SRL (AITEC) (Spain);
- FUNDACIO EURECAT (EURECAT) (Spain);
- WOMEN ENGAGE FOR A COMMON FUTURE EV (WECF) (Germany);
- UNIVERSIDAD COMPLUTENSE DE MADRID (UCM) (Spain);
- SZEKELY FAMILY & CO. NONPROFIT KORLATOLT FELELOSSEGU TARSASAG (SFC) (Hungary);
- FONDAZIONE GIACOMO BRODOLINI SRL SB (FGB) (Italy);
- UNINOVA-INSTITUTO DE DESENVOLVIMENTO DE NOVAS TECNOLOGIAS-ASSOCIACAO (UNINOVA) (Portugal).

3. Purpose of the research

FITTER-EU aims to contribute to existing research on the origins, dynamics and determinants of inequalities and enable anticipatory governance to support a fair and inclusive twin transition in Europe. The project innovates by the formulation and development of an ecosystem that will enable policymakers to predict which social groups may be at risk of being adversely affected by twin transition policies under different

scenarios, with the option to simulate the implementation of these policies to assess inequalities and social exclusion risks.

To achieve its objectives, FITTER-EU is collecting both quantitative and qualitative data on inequalities that may be reinforced, widened, or newly created as a result of twin transition policies (i.e., Green + Digital).

4. Participation in the project and right to withdraw

Your entry of personal data into the Stakeholder Database is entirely voluntary. You may withdraw your consent and opt out at any time, and you may do so without any explanation and with no consequences for your relation to your institution or to FITTER-EU project partners. You have the right to retract the data you have provided (or remove portions of them) at any time, including your personal data. If you wish to exercise your right to withdraw, you can contact the project coordinator, with no need to explain why you want to withdraw. Some activities we currently have planned are:

5. Procedure

By signing up to the FITTER-EU Stakeholder Database, you may be contacted by researchers as part of the project, inviting you to participate in the project's stakeholder engagement activities. These activities could include workshops, focus groups, interviews, surveys, hackathons, as well as other events and activities. Signing up to the Stakeholder Database does not guarantee that you will be contacted to take part in project activities. If you are contacted and invited to an activity, you will be free to decide whether to participate or not.

6. Data protection

All activities carried out in the FITTER-EU project comply with General Data Protection Regulation (EU) 2016/679. All personal data collected will be kept secure and no personal data will be kept for longer than necessary for the purposes for which they were collected. Personal data which are no longer required for the purposes of the project will be deleted at the end of the project, at the very latest.

7. Anonymity and confidentiality

The information gathered through the form will be kept in the strictest confidentiality and will be safely stored on a safe file GDPR certified repository owned and managed by the coordinator (TU Dublin) and accessible by log-in only by project partners. The file repository is Microsoft Teams file repository, whose GDPR compliance is described [here](#).

Team members from the consortium partners only will have access to the data. Your personal data will be kept confidential and will not be disclosed to any other individual or third parties.

The results of any activities you choose to participate in will be disseminated for scientific purposes, in compliance with the ethical standards of the scientific community. Reports or any scientific outcomes related to this project will not contain any personal information and you will not be identified or identifiable.

8. Possible risks, disadvantages or discomfort

Your participation in this research does not present any physical or psychological risks, risk of disadvantages, and does not intend any discomfort. Should you feel distressed at any point during project activities, please inform consortium members immediately and feel free to interrupt your participation and withdraw and/or ask for cancellation of your contact/data.

9. Project Results

We will keep you informed about the results of the study. These will be reported to the European Commission, as well as in research articles and on digital platforms. Data will only be made publicly

available when it is possible to protect the privacy and anonymity of individuals. Some research results and news items will be posted on the project website (<http://fitter-project.eu>).

10. Questions and claims

If you have any questions about the research or if you would like to make a complaint on how you have been treated during your participation in this project, please contact the FITTER-EU Ethics Manager or the Project Coordinator (contacts below).

11. Organisation and Research Funding

FITTER-EU has received funding from the European Union's Horizon Europe research and innovation programme under Grant Agreement No 101132546.

12. Ethical Review Project

The project has passed the ethics review process set by the European Commission.

13. Contact for more information

Project coordinator: Sara Clavero, sara.clavero@tudublin.ie

FITTER-EU Ethics Manager: Zoltán Székely, zoltan@szekely.family

Annex N° 4: Templates for stakeholder engagement

In this annex, a series of templates to be used as a reference for the top-down stakeholder engagement procedures —see Chapter N° 4— are detailed: (i) a template to draft a presentation and invitation to a FITTER-EU participatory event, (ii) registration form template, (iii) informed consent form template, (iv) ex post feedback survey for participants to FITTER-EU events, (v) template of a follow-up email to participants.

- **Information sheet: template to draft a presentation and invitation to a FITTER-EU participatory event**

Dear *[insert name]*,

We are pleased to extend an invitation for your participation in *[insert description of the activity]* as part of the FITTER-EU project *[insert link of the project's webpage]*, which will take place on *[insert details on the location and date]*.

About FITTER-EU project

The FITTER-EU project is dedicated to researching the origins, dynamics, and determinants of inequalities related to the twin transition, giving a particular Focus on the six countries of the project: Germany, Hungary, Ireland, Italy, Portugal and Spain. The project studies the effects of the transitions across four sectors —Energy, Transport, Building & Housing, Food & Agriculture — and the policy responses that support decision-making for a just twin transition in Europe. The final outcome of the project is the development of the FITTER Ecosystem, a highly interactive and gamified digital platform powered by an Intelligent Predictive Decision Support System. This platform will inform and provide inputs to policymakers regarding the effects of the twin transitions in prioritised population groups at-risk of suffering their adverse effects.

Details on *[name of the activity/event]*

The activity *[insert name/details]* is aimed at *[include specific objectives]*. Your expertise and input will be invaluable to the success of this task. Below, you can find more details on the activity.

- **Objectives** *[insert specific objectives]*.
- **Duration:** *[insert timeframe and milestones]*.
- **Format:** *[insert format of the activity, detailing the platform in case of an online event or the venue in case of an in-presence event]*
- **Target Participants:** *[insert target participants considering the stakeholder categories of project]*
- **Expected Outcomes:** *[insert what participants can expect to achieve or contribute concretely as part of the activity]*

Why Participate?

By contributing to this task, you will: *[Modify, select, or include other bullet points related to the specific activity you are implementing]*

- Network with key stakeholders, including policymakers, civil society organisations, trade unions, employer's organisations, business angels and actors from the private sector and public institutions.
- Gain visibility and contribute to cutting-edge research on inequality and social justice during the green and digital transitions.
- Play a critical role in shaping policy recommendations that ensure an equitable twin transition
- Provide input that will directly influence the development of the FITTER Ecosystem, contributing to the decision-making to tackle the adverse social impacts of the twin transition.
- Test the FITTER Ecosystem and contribute to their validation.
-

To confirm your participation

If you are interested in contributing to this activity, please express your interest by filling out the registration form *[insert link of the registration form, see next section of the Annex]*. Once we receive your confirmation, we will provide more details regarding your participation and next steps.

Your engagement is key to the success of this project, and we greatly value your contribution to ensuring a just and inclusive transition for all.

Thank you for considering this opportunity. We look forward to collaborating with you.

Best regards,

[Name of the project partner, organisation and email]

- **Registration form template to FITTER-EU events**

Use the following online form as a template:

- [Link](#) to send the registration form to participants.
- Editable [link](#) to customise the form to the participatory activity.

- **Informed consent form on participation to FITTER-EU events**



Research information for informed consent (*name of the activity*)

FITTER-EU: Fair and inclusive twin transitions for a stronger social Europe

1. Introduction

I am contacting you to invite you to participate in [*insert activity name*] for a study organised within FITTER-EU, a project funded by the European Commission under the Horizon Europe Framework Programme. Before you decide whether or not to take part, it is important for you to understand why the research is being done and what it will involve. Please take time to read the following information carefully.

2. Purpose of the research

FITTER-EU aims to contribute to existing research on the origins, dynamics and determinants of inequalities and enable anticipatory governance to support a fair and inclusive twin transition in Europe. The project innovates by the formulation and development of an ecosystem that will enable policymakers to predict which social groups may be at risk of being adversely affected by twin transition policies under different scenarios, with the option to simulate the implementation of these policies to assess inequalities and social exclusion risks.

To achieve its objectives, FITTER-EU is collecting both quantitative and qualitative data on inequalities that may be reinforced, widened, or newly created as a result of twin transition policies (i.e., Green + Digital). You have been invited to participate because you were identified as an expert in [*include details*]. Your experience is important to our qualitative research. The researchers will use this activity to [*include details*].

3. Participation in the project and right to withdraw

Your participation is entirely voluntary. You will be invited to read this information sheet and sign it to indicate consent, in print or electronically. If you consent to participate, you will take part in *[details name of the activity and expected duration]*. You may withdraw your consent and opt out at any time, and you may do so without any explanation and with no consequences for your relation to your institution or to FITTER-EU project partners. During the implementation of the activity, you do not have to answer any questions you do not wish to. You have the right to retract the data you have provided (or remove portions of them) at any time, including your personal data. If you wish to exercise your right to withdraw after the focus group you can contact the principal investigator, with no need to explain why you want to withdraw.

4. Procedure *[adapt to procedure: audio or video recording, note-taking, other.....]*

This research will involve your participation in *[include details]* with other experts in the area of *[include details]*. The activity will be recorded for the purposes of analysis. The information recorded is confidential. Your personal data will be anonymised and no one else except the project research team at *[include details]* will have access to the recording, which will be stored in a safe, password protected, place. Once the activity is finished, portions of the recording will be transcribed, and summarised in a report. Your name will not be mentioned in this report. The data will be processed so that unauthorised persons cannot access it.

5. Data protection

All activities carried out in the FITTER-EU project comply with General Data Protection Regulation (EU) 2016/679. All personal data collected will be kept secure and no personal data will be kept for longer than necessary for the purposes for which they were collected. Personal data which are no longer required for the purposes of the project will be deleted, and all recordings will be deleted at the end of the project, at the very latest.

6. Anonymity and confidentiality

The information gathered through the form will be kept in the strictest confidentiality and will be safely stored on a safe file GDPR certified repository owned and managed by the coordinator (TU Dublin) and accessible by log-in only by project partners. The file repository is Microsoft Teams file repository, whose GDPR compliance is described [here](#) *[adapt this information if you are going to storage this information in a different repository]*.

Team members from the consortium partners only will have access to the data. Your personal data will be kept confidential and will not be disclosed to any other individual or third parties.

The results of any activities you choose to participate in will be disseminated for scientific purposes, in compliance with the ethical standards of the scientific community. Reports or any scientific outcomes related to this project will not contain any personal information and you will not be identified or identifiable.

7. Possible risks, disadvantages or discomfort

Your participation in this research does not present any physical or psychological risks, risk of disadvantages, and does not intend any discomfort.

8. Project Results

We will keep you informed about the results of the study. These will be reported to the European Commission, as well as in research articles and on digital platforms. Data will only be made publicly

available when it is possible to protect the privacy and anonymity of individuals. Some research results and news items will be posted on the project website (<http://fitter-project.eu>)

9. Questions and claims

If you have any questions about the research or if you would like to make a complaint on how you have been treated during your participation in this project, please contact the principal investigator, who will process it quickly and with utmost care. If you are unsatisfied with the way your complaint is being handled, you can contact the FITTER-EU Ethics Manager or the Project Coordinator (contacts below).

10. Organisation and Research Funding

FITTER-EU has received funding from the European Union's Horizon Europe research and innovation programme under grant agreement No 101132546

11. Ethical Review Project

The project has passed the ethics review process set by the European Commission.

12. Contact for more information

Principal Investigator: (of the institution responsible for this research)

Project coordinator: Sara Clavero, sara.clavero@tudublin.ie

FITTER-EU Ethics Manager: Zoltán Székely, zoltan@szekely.family

PARTICIPANT STATEMENT

First, I manifest that I have been invited to participate in the FITTER-EU project and that I consent voluntarily to participate in a focus group for the purposes of this study.

Second, I confirm that I have read the foregoing information, and I have been properly informed. I have received and understood the information provided about the content, purposes and benefits of this project. I have had the opportunity to ask questions about it and any questions I have asked, have been answered to my satisfaction. Therefore, I am aware that my participation is voluntary and that I am free to withdraw at any time without giving any reason and without the risk of any negative consequences. I understand that the recording of the focus group *[adapt/modify based on procedure]* will be kept by the FITTER-EU project research team, and destroyed at the end of the project, at the latest. I consent to the information I provide being used for research purposes.

Name of Participant _____

Signature of Participant _____

Date (Day/month/year) _____

STATEMENT BY THE RESEARCHER/PERSON TAKING CONSENT

First, I have provided a copy of the research project information sheet to the prospective participant, and to the best of my ability have made sure that the participant understands the information provided about the content, purposes and benefits of this project. A copy of this informed consent form has been provided to the participant.

Second, I confirm that the participant has been given an opportunity to ask questions about the research and all the questions asked by the participant have been answered correctly and to the best of my ability. I confirm that the consent is informed and has been given freely and voluntarily.

Name of Researcher _____

Signature of Researcher

Date (Day/month/year) _____

- **Ex post feedback survey for participants to FITTER-EU events**

Use the following online form as a template:

- [Link](#) to send the questionnaire to participants.
- Editable [link](#) to customise the form to the participatory activity.

Before customising the online form to your activities, **kindly consider creating a copy to avoid changes in the template.**

- **Template for a follow-up e-mail to participants**

Dear *[insert name]*,

I hope this message finds you well. On behalf of the FITTER-EU project team, I would like to extend our sincere gratitude for your valuable contribution to *[insert activity name]*, which took place on *[insert date]*.

Your insights and expertise played a significant role in advancing our efforts to *[insert task objectives]*.

If you haven't had the chance to do so yet, we invite you to reply to the following [online questionnaire](#) *[see previous section to customise the questionnaire]*. Your feedback will help us enhance the quality and effectiveness of future activities.

We will keep you updated on the progress of this task and any upcoming activities that may be of interest to you. Do not hesitate to reach us in case of any further thoughts or feedback. Your ongoing input is always welcome.

Thank you once again for your time and collaboration.

Best regards,

[Name of the project partner, organisation and email]

Annex N° 5: Criteria for the inclusiveness checks

The following table presents the criteria for the periodic inclusiveness —see section 4.5.2: Periodic internal inclusiveness checks (task ownership: each partner in charge of national case studies, coordination by SV and TFC)—.

Table 14: Inclusiveness checklist criteria

Criteria	Indicator	Notes	Score (0-5)
Diversity of Sectors Represented	Representation from public, private, academia, civil society, business angels	Please specify how many of the participating stakeholders were from the various categories/types, in the addressed timeframe	0: One sector, 5: Balanced representation
Cross-disciplinary Representation	Representation from both digital and green transition fields	Please briefly specify to what extent participants had knowledge/expertise and/or are directly impacted by green and/or digital transitions policies	0: One field, 5: Balanced representation
Geographical Representation	Representation from diverse regions/ central and periphery areas (or administrative levels for policy makers)	Please add a few explanatory notes	0: One region, 5: Diverse regions/administrative levels
Social Inclusion (Marginalised/disadvantaged groups, minorities, social movements. CSOs, non-mainstream actors with critical approaches etc.)	Involvement of marginalised groups, SMEs	Please specify out of the participating stakeholders what share was representing/from the mentioned categories	0: No marginalised groups, 5: Strong involvement
Digital Accessibility (Platforms, Tools)	Platforms fully accessible, including for people with disabilities, or alternatives offered	Please add a few explanatory notes	0: No accessibility, 5: Fully accessible platforms
Inclusive language	Inclusive language used in promoting and during the events (LGBTQIA* sensitive use of pronouns, avoidance of stereotypes based on gender, race,	Please briefly describe in few words how you addressed this	

Criteria	Indicator	Notes	Score (0-5)
	social class, sexual orientation or disability etc, avoidance of tokenism)		
Knowledge accessibility	Terminology in use fully understandable by all participants, minimising or translating/making technical jargon accessible		0: No accessibility, 5: Fully accessible platforms
Virtual and In-person Engagement	Virtual engagement available upon request to facilitate participants		0: One option, 5: Virtual and in-person
Balanced Power Dynamics	Equal contribution opportunities for all stakeholders, encouragement offered to those less acquainted with research/policy/innovation settings	Please briefly describe in few words how you addressed this	0: Dominated by a few, 5: Equal opportunities/participation
Support to Resource-constrained Stakeholders	Forms of concrete support foreseen for marginalised stakeholders, even if on a case-by-case basis (coverage of travel costs, vouchers for carers, baby sitting available, etc.)	Please briefly describe in few words how you addressed this	0: No compensation, 5: Fair compensation
Care -sensitive timing/scheduling	Scheduling meetings to address needs of participants with care responsibility and facilitate work life balance (avoid too early/late hours, weekends etc.)	Please briefly describe in few words how you addressed this	0: No care-sensitive scheduling, 5: Fully care-sensitive scheduling
Opportunities for Stakeholder Feedback	Encouraging feedback, indicating contact persons and keeping open channels for communication	Please specify if feedback surveys were administered and follow up emails sent	0: No feedback, 5: Regular feedback and adaptations

Criteria	Indicator	Notes	Score (0-5)
Gender Representation	Equal gender representation among participants and gender balance in discussions	Please indicate the average M/F ratio of participants in the related timeframe	0: No attention, 5: Equal gender representation
Gender/intersectionality sensitive Analysis	Gender-intersectionality sensitive analysis/disaggregated data integrated into presentations/ discussions		0: No gender considerations, 5: Comprehensive analysis
Data Privacy and Security	Compliance with GDPR and data transparency		0: No compliance, 5: Full compliance with GDPR
Capacity Building Opportunities	Capacity-building programs for stakeholders		0: No programs, 5: Regular capacity-building programs
Adaptability of Engagement Process	Processes adapted based on stakeholder feedback	Please briefly specify if/ how based on the received feedback you plan to adapt your engagement process	0: No adaptation, 5: Processes adapted regularly

